

9th International Conference on
**Design, Simulation, Manufacturing:
The Innovation Exchange**

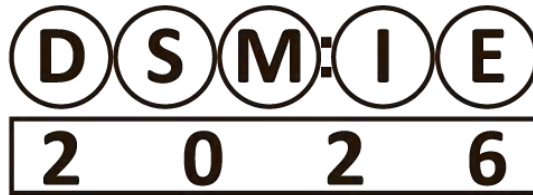
June 2-5, 2026 | Almaty, Kazakhstan

Book of Abstracts



Together we can do more for science, technology, engineering, and education.

International Association for Technological Development and Innovations



**9th International Conference on
Design, Simulation, Manufacturing:
The Innovation Exchange
(DSMIE-2026)**

June 2-5, 2026 | Almaty, Kazakhstan

Book of Abstracts

Almaty
2026

Editors:

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*Recommended by the Coordination Board of the
International Association for Technological Development and Innovations
(Protocol No. 5, January 12, 2026).*

The content of this book is based on the 8th International Conference on Design, Simulation, Manufacturing: The Innovation Exchange (DSMIE-2026), held on June 2-5, 2026, in Almaty, Kazakhstan. This book reports on advances in manufacturing, mechanical, and materials engineering, as well as industrial management. This book provides academics and professionals with extensive information on trends, technologies, challenges, and practice-oriented experience in all the areas mentioned above.

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Welcome Message

The 9th International Conference on Design, Simulation, Manufacturing: The Innovation Exchange (DSMIE-2026), held in Almaty, Kazakhstan, on June 2-5, 2026. The conference was organized by the International Association for Technological Development and Innovations, NNEF Public Foundation, International IT University, and Sumy State University, in partnership with Poznan University of Technology (Poland), Kielce University of Technology (Poland), Technical University of Kosice (Slovak Republic), Association for Promoting Innovative Technologies – Innovative FET (Croatia), and Society for Robotics of Bosnia and Herzegovina (Bosnia and Herzegovina).

DSMIE-2026 received 154 contributions from 35 countries worldwide. After a two-stage single-blind review, the Program Committee accepted 72 papers written by 309 authors from 31 countries. Thank you very much to the authors for their contribution. These papers are published in the present book, achieving an acceptance rate of about 47%. Extended versions of selected best papers will be published in the following scientific journals: International Journal of Advanced Manufacturing Technology (Springer), Production & Manufacturing Research (Taylor and Francis), and Scientific Reports (Springer Nature).

We want to thank the members of the Program Committee and the invited external reviewers for their efforts and expertise in reviewing, without which it would be impossible to maintain the high standards of peer-reviewed papers. 87 Program Committee members and 14 invited external reviewers devoted their time and energy to peer-reviewing manuscripts. Our reviewers come from around the world, representing 18 countries, and are affiliated with 48 institutions.

Thank you very much to the keynote speakers: Prof. Raul Campilho (Polytechnic of Porto, Portugal), Prof. Khamis Essa (University of Birmingham, UK), Prof. Abhinav Dhall (Monash University, Australia), Prof. Filip Gorski (Poznan University of Technology, Poland), and Prof. Torbjorn Netland (ETH Zurich, Switzerland).

We are deeply grateful to the invited speakers who contributed to the Business to Science session: Iurii Gaidaienko (Yalong Intelligent Equipment Group Co. Ltd., China).

We sincerely appreciate the invited lecturers of the DSMIE Workshop, Dr. Bohdan Haidabrus (Accenture Baltics / Riga Technical University, Latvia) and Pawel Czyżewski (Poznan University of Technology, Poland).

Thank you very much to Co-Chairs, Dr. Bohdan Haidabrus (Riga Technical University, Latvia), Prof. Oleksandr Liaposhchenko (Sumy State University, Ukraine), and Dr. Justyna Trojanowska (Poznan University of Technology, Poland)

for their coordination with the program committee members and for the ensuring high quality of research papers, Prof. Raul Campilho (Polytechnic of Porto, Portugal) and Prof. Francisco J.G. Silva (Polytechnic of Porto, Portugal) for support with the peer-review process and selecting papers for conference proceedings, Dr. Khrystyna Berladir (Sumy State University, Ukraine) for contribution to the editorial management of the conference and for maintaining effective communication with authors, Mrs. Yerkezhan Amangozhayeva (International IT University, Kazakhstan) for organizational issues, and Mrs. Kateryna Ivanova (International Association for Technological Development and Innovations, Ukraine) for contribution to the development and promotion of the DSMIE Conference Series brand, as well as for the preparation, visualization, and dissemination of promotional materials.

We appreciate the partnership with Springer Nature, Turnitin, and the International Innovation Foundation for their essential support during conference preparation.

Thank you very much to the DSMIE Team. Their involvement and hard work were crucial to the success of the conference.

DSMIE's motto is ***"Together we can do more for science, technology, engineering, and education"***.

Vitalii Ivanov,
General Chair

Kateryna Kolesnikova,
General Co-Chair, Local Chair

About DSMIE-2026

The 9th International Conference on Design, Simulation, Manufacturing: The Innovation Exchange (DSMIE-2026) is the international forum for fundamental and applied research and industrial applications in engineering.

The conference focuses on research challenges in the fields of Manufacturing Engineering, Materials Engineering, Mechanical Engineering, and Industrial Management, addressing current and future trends in design approaches, simulation techniques, and manufacturing technologies, highlighting the growing role of smart manufacturing systems, artificial intelligence, standards-based integration, and innovations implementation in the transition to sustainable, human-centric, and resilient engineering solutions.

DSMIE brings together researchers from academic institutions, leading industrial companies, and government laboratories worldwide to promote and popularize the scientific fundamentals of engineering.

The official language of the conference is English.

History

The history of the DSMIE Conference Series started in 2018. Since then, DSMIE has become an annual event, aiming to foster exchange of ideas, spread innovation, and attract more participants and institutions. DSMIE Conference Series changed several locations in Ukraine (2018–2021) and was successful in Europe (Poland, Slovak Republic, Czech Republic, and Portugal). Now, the 9th edition of the DSMIE Conference Series has been held in Central Asia (Kazakhstan).

DSMIE-2018 | Sumy, Ukraine

DSMIE-2019 | Lutsk, Ukraine

DSMIE-2020 | Kharkiv, Ukraine

DSMIE-2021 | Lviv, Ukraine

DSMIE-2022 | Poznan, Poland

DSMIE-2023 | High Tatras, Slovak Republic

DSMIE-2024 | Pilsen, Czech Republic

DSMIE-2025 | Porto, Portugal

DSMIE-2026 | Almaty, Kazakhstan

We are proud of our achievements over the past years. It is the result of a reliable, long-term partnership with all our partners and host institutions of the DSMIE Conference Series, as well as the significant contributions of participants.

Organizers



Together we can do more for science, technology, engineering, and education.

International Association for Technological Development and Innovations

The International Association for Technological Development and Innovations (IATDI) is a non-profit organization that promotes technological development and innovation across various industries and sectors. The association is committed to advancing research and development efforts in technology, engineering, and applied sciences and fostering collaboration and knowledge sharing among its members. IATDI aims to bridge the gap between academia and industry and create a conducive environment for developing and adopting new technologies through its various programs and initiatives. The association also provides a platform for its members to network, collaborate, and share knowledge, fostering a community of like-minded individuals committed to advancing technology and innovation.

✉ 5/30, M. Lushpy Ave., Office 29, 40035, Sumy, Ukraine

🌐 <http://iatdi.org/>

NNEF Educational Foundation

NNEF Educational Foundation is a non-profit public foundation established at the initiative of the First President of the Republic of Kazakhstan. Headquartered in Almaty, Kazakhstan, the Foundation is dedicated to building a comprehensive and continuous educational ecosystem – from early childhood through higher education and lifelong learning. Operating under the motto "Bridge to the Future," NNEF pursues the development of innovative, high-quality education across all levels of learning.

Over the course of its activity, the Foundation has built one of Kazakhstan's most extensive private educational clusters, which encompasses three innovative Miras kindergartens in Astana, Almaty, and Atyrau; four schools including two international Miras schools, the International School of Almaty, and the Arystan Specialized Lyceum; and an Alliance of five universities of science and technology –

the International University of Information Technology (IITU), Astana IT University (AITU), Kazakh-British Technical University (KBTU), Almaty University of Power Engineering and Telecommunications (AUPET), and the University of International Business (UIB). The Foundation also supports a social initiative through SOS Children's Villages Kazakhstan. Together, these institutions serve over 31,000 students across more than 163,000 square meters of educational space, with a staff of over 6,000.

NNEF positions itself as a unique ecosystem of continuous education for individuals from age 2 to the silver generation, fostering academic excellence, technological innovation, and social responsibility throughout Kazakhstan.

✉ 64A, Iskenderov St., Almaty, Kazakhstan

🌐 <https://foundation.kz/>

International Information Technology University

The International Information Technology University (IITU) is a higher educational institution established in Almaty, Kazakhstan, at the initiative of the President of the Republic of Kazakhstan. Part of the NNEF Educational Foundation's cluster, IITU is dedicated to training a new generation of qualified, internationally recognized ICT specialists capable of meeting the demands of Kazakhstan's information and communication technology industry and contributing to the country's integration into the global digital economy.

IITU is the only university in Kazakhstan to have received ASIIN institutional international accreditation, confirming its compliance with the standards of the Agency for Accreditation of Educational Programs in Engineering, Computer Science, Natural Sciences and Mathematics (Germany). The university offers bachelor's, master's, PhD, and MBA programs across two faculties – the Faculty of Computer Technology and CyberSecurity, and the Faculty of Business, Media and Management – enrolling more than 5,000 students. IITU has consistently ranked among the top three universities in Kazakhstan by the National Chamber of Entrepreneurs "Atameken," placing first in programs such as Information Systems and Electronic Journalism.

The university maintains academic partnerships with over 70 universities across North America, Europe, East Asia, and the CIS, offering exchange programs, double degrees, and research collaboration opportunities. IITU was the first university in Kazakhstan and Central Asia to partner with Coursera for Campus, integrating blended learning into its programs. A Cybersecurity Competence Center and a dedicated military department for cybersecurity training further distinguish IITU as a leading institution in this field. With a graduate employment rate consistently at

around 90%, IITU combines rigorous academic training with active student life, entrepreneurship, and international experience.

✉ 34/1 Manasa St., Bostandyk district, Almaty, Republic of Kazakhstan

🌐 <https://iitu.edu.kz/>

Sumy State University

Sumy State University (SumDU) is a leading classical university in Ukraine. It cooperates with more than 300 partners from 55 countries worldwide. SumDU is a reliable partner for joint projects in the framework of international grant programs of the EU, the United Nations Development Program, NATO, DAAD, American Councils, the British Council, NAWA, JICA, the World Bank, bilateral scientific and research projects, and grants from private foundations. The University accomplishes more than 300 grants annually.

✉ 116, Kharkivska St., 40007, Sumy, Ukraine

🌐 <https://sumdu.edu.ua/>

Partners



Sponsors

Visit Almaty



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Justyna Żywiołek	Czestochowa University of Technology, Poland

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Conference Topics

Manufacturing Engineering

- Product Design and Advanced Manufacturing Processes
- Digital Twins, Intelligent Manufacturing Systems, IIoT
- Automation, Robotics, and Assembly Systems
- Smart Manufacturing and Industry 4.0/5.0 Strategies
- Decision-Making in Manufacturing Control and Optimization
- Information Management Systems and Standards-Based Integration
- Sustainable and Resilient Manufacturing
- Engineering Education and Professional Growth

Materials Engineering

- Methods and Technologies for Additive Manufacturing
- Advanced Materials for Industrial Applications
- Materials Modeling, Simulation, and Optimization

Mechanical Engineering

- Mechanics of Fluids, Solids, and Structures
- Dynamics, Acoustics, and Vibrations
- Process Technology and Engineering
- Numerical Simulations of Coupled Problems
- Thermal and Chemical Engineering

Industrial Management

- Increasing Productivity
- Production Scheduling and Simulation
- Advances in Quality Management
- Lean Manufacturing
- Continuous Improvement Methodologies and Case Studies
- Advanced Maintenance Approaches
- Safety at Work
- Industrial Sustainability
- Logistics and Supply Chain Engineering
- Novel Statistical Approaches in Manufacturing and Applications
- Agile Manufacturing

Publishing Opportunities

Conference Proceedings

Full papers of selected contributions of DSMIE-2026 were published in three volumes in the book **“Advances in Design, Simulation and Manufacturing IX”**. It belongs to the Lecture Notes in Mechanical Engineering series (ISSN 2195-4356). The books of this series are published by Springer Nature, indexed by Scopus and EI Compindex, and submitted to the Web of Science Core Collection.



Volume 1 – Manufacturing Processes and Intelligent Production Systems

Editors:

- Vitalii Ivanov, Sumy State University, Ukraine
- Kateryna Kolesnikova, International IT University, Kazakhstan
- Justyna Trojanowska, Poznan University of Technology, Poland

Volume 2 – Design, Simulation, and Energy Systems

Editors:

- Vitalii Ivanov, Sumy State University, Ukraine
- Raul D. Campilho, Polytechnic of Porto, Portugal
- Francisco J.G. Silva, Polytechnic of Porto, Portugal

Volume 3 - Materials Engineering, Digital Transformation, and Human Factors in Engineering

Editors:

- Vitalii Ivanov, Sumy State University, Ukraine
- Kateryna Kolesnikova, International IT University, Kazakhstan
- Bohdan Haidabrus, Riga Technical University, Latvia

To read the full papers, please visit the official webpage of the Publisher via the following link: <https://link.springer.com/conference/dsmie> or DSMIE's website: <https://dsmie.sumdu.edu.ua/schedule/proceedings.html>.

Partner Journals

Extended versions of the best papers, presented at DSMIE-2026, will be considered for publication in the special issues and partner journals, subject to further review and article processing charges (if applicable):

- The International Journal of Advanced Manufacturing Technology (Springer, ISSN 0268-3768), <https://link.springer.com/journal/170>;
- Production & Manufacturing Research (Taylor and Francis, ISSN 2169-3277), <https://www.tandfonline.com/journals/tpmr20>;
- Scientific Reports (Springer Nature, ISSN 2045-2322), <https://www.nature.com/srep/>.

Venue

Almaty, formerly Alma-Ata, is the largest city in Kazakhstan, with a population exceeding 2 million in its metropolitan area. Located in the foothills of the Trans-Ili Alatau mountains in southern Kazakhstan, Almaty stands as a pivotal center of culture, commerce, finance, and innovation. The city is nestled at an elevation of 700–900 meters (2,300–3,000 feet), with the Big Almaty and Small Almaty rivers running through it, originating from the surrounding mountains and flowing into the plains. Almaty is the second-largest city in Central Asia.

Almaty served as the capital of Kazakhstan from 1929 to 1997 during the Soviet era, and after independence from 1991 until 1997, when the capital was relocated to Akmola (now Astana). Despite no longer being the capital, Almaty remains the most cosmopolitan and influential city in Kazakhstan, often regarded as the nation's cultural and financial heart, and is colloquially referred to as the "Southern Capital". It is classified as a city of republican significance, granting it autonomy from regional governance. The city is divided into eight administrative districts within the framework of its republican significance.



Almaty has hosted international events, including the Alma-Ata Primary Healthcare Conference in 1978, which shaped global public health policy, the 2011 Asian Winter Games, and the 2017 Winter Universiade. The city was also a contender to

host the 2022 Winter Olympics, further solidifying its position as an international hub for winter sports, but ultimately lost the bid to Beijing.

Almaty is served by an extensive transportation network, including Almaty International Airport, the busiest in Kazakhstan and Central Asia, handling over 9.5 million passengers annually. The city is also served by the Almaty Metro, opened in 2011. Additionally, buses and trolleybuses contribute to the city's comprehensive public transport system.

Almaty's proximity to natural attractions such as the Ile-Alatau National Park, with its diverse ecosystems and scenic trails, and Big Almaty Lake, makes it a popular destination for outdoor enthusiasts. Additionally, the city's close proximity to Medeu, the highest-altitude ice rink in the world, and Shymbulak ski resort, contributes to Almaty's reputation as a major tourism hub for adventure and outdoor activities.

Agenda



Agenda online

Day 1 | June 2, 2026

- | | |
|------------------------------------|------------------------------|
| 12 ⁰⁰ –15 ⁰⁰ | Industry Tour – Eltex Alatau |
| 15 ⁰⁰ –17 ⁰⁰ | City Tour |

Day 2 | June 3, 2026

- | | |
|------------------------------------|---|
| 09 ⁰⁰ –10 ⁰⁰ | Reception & Registration |
| 10 ⁰⁰ –10 ³⁰ | Opening Ceremony |
| 10 ³⁰ –12 ³⁰ | Keynote Session I |
| 12 ³⁰ –14 ⁰⁰ | Lunch |
| 14 ⁰⁰ –15 ³⁰ | Session 1. Industrial Management & Sustainable Manufacturing (Room 1) |
| 14 ⁰⁰ –15 ³⁰ | Session 2. Computational & Digital Methods in Engineering (Room 2) |
| 15 ³⁰ –16 ⁰⁰ | Coffee Break |
| 16 ⁰⁰ –17 ³⁰ | Session 3. Intelligent Monitoring & Engineering Systems (Room 1) |
| 16 ⁰⁰ –17 ³⁰ | Session 4. Industrial Digitalization & Data-Driven Systems (Room 2) |
| 18 ³⁰ –21 ⁰⁰ | Welcome Dinner |

Agenda (continuous)



Agenda online

Day 3 | June 4, 2026

08 ⁰⁰ –09 ⁰⁰	Morning Coffee
08 ⁰⁰ –17 ⁰⁰	Video Presentations
09 ⁰⁰ –11 ⁰⁰	Keynote Session II
11 ⁰⁰ –11 ³⁰	Coffee Break
11 ³⁰ –12 ³⁰	Business to Science Session
12 ³⁰ –14 ⁰⁰	Lunch
14 ⁰⁰ –15 ⁴⁵	Session 5. Advanced Manufacturing (Room 1)
14 ⁰⁰ –15 ⁴⁵	Session 6. Advanced Manufacturing Processes (Room 2 – Online)
15 ⁴⁵ –16 ¹⁵	Coffee Break
16 ¹⁵ –18 ⁰⁰	DSMIE Workshop
19 ⁰⁰ –22 ⁰⁰	Awards Ceremony & Gala Dinner

Day 4 | June 5, 2026

09 ⁴⁵ –10 ⁰⁰	Brief Orientation
10 ⁰⁰ –18 ⁰⁰	Cultural Program (Outside of Almaty)
18 ⁰⁰ –20 ⁰⁰	Return to Almaty

Day 1 | June 2, 2026

12⁰⁰–15⁰⁰ **INDUSTRIAL TOUR (Eltex Alatau)**
15⁰⁰–17⁰⁰ **CITY TOUR**

Day 2 | June 3, 2026

Time	Room 1	Room 2
09 ⁰⁰ –10 ⁰⁰	RECEPTION & REGISTRATION	
10 ⁰⁰ –10 ³⁰	OPENING CEREMONY	
10 ³⁰ –12 ³⁰	KEYNOTE SESSION I Session Chair: Vitalii IVANOV, Sumy State University, Ukraine	
10 ³⁰ –11 ¹⁰	Next-Generation Product Development: Integrating Human-Centered Design, Circularity, and Simulation for Sustainable Manufacturing Keynote Speaker: Raul CAMPILHO, Polytechnic of Porto, Portugal	
11 ¹⁰ –11 ⁵⁰	Shaping the Future: Advanced Netshape Technologies for High Value Manufacturing Keynote Speaker: Khamis ESSA, University of Birmingham, United Kingdom	
11 ⁵⁰ –12 ³⁰	Deepfakes and Misinformation in the Age of Industry 4.0 Keynote Speaker: Abhinav DHALL, Monash University, Australia	
12 ³⁰ –14 ⁰⁰	LUNCH	
14 ⁰⁰ –15 ³⁰	SESSION 1: INDUSTRIAL MANAGEMENT & SUSTAINABLE TRANSFORMATION Session Chair: Malgorzata JASIULEWICZ-KACZMAREK, Poznan University of Technology, Poland	SESSION 2: COMPUTATIONAL & DIGITAL METHODS IN ENGINEERING Session Chair: Dana TSOY, International IT University, Kazakhstan
14 ⁰⁰ –14 ¹⁵	Product Management Capabilities in a Data Programme Implementation <u>Bohdan Haidabrus</u> , Jānis Grabis, and Eugen Druzhinin	Implementation of Successive Overrelaxation Method for Solution of Elliptic Boundary-Value Problem <u>Bakhyt Alipova</u> and James McDonough

Time	Room 1	Room 2
14 ¹⁵ –14 ³⁰	Evaluation of investment attractiveness in the context of Industry 4.0/5.0 transformation <u>Iryna Bashynska</u> , Olena Zhuk, Ihor Bashynskiy, and Maksym Pastukhov	Fast Neuro-Symbolic Constraint Solving for Knowledge Extraction via NP-Complete Subset Sum Models <u>Anel Auyezova</u> , Aigerim Aitim, Bakhtgery Sinchev, and Waldemar Wójcik
14 ³⁰ –14 ⁴⁵	Market Readiness for Data Science as a Service: An Empirical Multi-Phase Process Assessment Sebastian Podschlapp and <u>Philipp Rosenberger</u>	A Cosserat Rod Approach to Static Analysis of Multi-Muscle Soft Manipulator Segments Oleksandr Sokolov, Vladyslav Andrusyshyn, Alexander Hošovský, <u>Pavlo Pomin</u> , and Khrystyna Berladir
14 ⁴⁵ –15 ⁰⁰	A Structured Investigation of Barriers to Reverse Logistics Implementation for Post-Consumer PET Bottles Umindu Liyanage, Nimantha T. Thalagala, <u>Amila Thibbotuwawa</u> , and Izabela Nielsen	QazNLP: A Morphology-Aware Kazakh NLP Toolkit with Structured Decoding for POS and NER <u>Aigerim Aitim</u> and Anel Auyezova
15 ⁰⁰ –15 ¹⁵	Navigating the Complexities of Responsible Sourcing in Global Supply Chains Buddhima Withanage, Nimantha T. Thalagala, and <u>H. Niles Perera</u>	Comparative Analysis of Cryptographic Algorithms for Securing IoT Systems Zhanerke Temirbekova and <u>Zhansaya Myrzakul</u>
15 ¹⁵ –15 ³⁰	Life Cycle Assessment Approach to Individualized 3D-Printed Orthopaedic and Prosthetic Devices Filip Gorski and <u>Yuliia Denysenko</u>	Optimized FHE Library Based on Pailler's Algorithm for Secure Computing in IoT Networks Aitulek Zhumakyn, Tolganay Chinibayeva, Nurdaulet Kabdygaliyev, <u>Zhanerke Temirbekova</u> , and Ualikhan Shokan
15 ³⁰ –16 ⁰⁰	COFFEE BREAK	

Time	Room 1	Room 2
16 ⁰⁰ –17 ³⁰	SESSION 3: INTELLIGENT MONITORING & ENGINEERING SYSTEMS Session Chair: Raul CAMPILHO, Polytechnic of Porto, Portugal	SESSION 4: INDUSTRIAL DIGITALIZATION & DATA-DRIVEN SYSTEMS Session Chair: Bohdan HAIDABRUS, Riga Technical University, Latvia
16 ⁰⁰ –16 ¹⁵	A Cognitive Digital Triplet Architecture for Anomaly Diagnosis and Operator-in-the-Loop Decision Support in Discrete Manufacturing <u>Slawomir Luscinski</u>	An Integrated Framework for Digital Advertisement and Fleet Capacity Optimization in Long-Haul Logistics Faten Afandi and <u>Chefi Triki</u>
16 ¹⁵ –16 ³⁰	Real-Time Navigation in Crowds: Using Optimized CCTV-Based Yolo for Robot Motion Planning Aidarkhan Kaimov, Talgat Kaiym, <u>Dastan Zhalmukhanbetov</u> , Amirzhan Myrzabayev, and Suleimen Kaimov	GenAI in Research Idea Lifecycle: Conceptual Framework <u>Artem Artyukhov</u> , Nadiia Artyukhova, and Dmytro Chumachenko
16 ³⁰ –16 ⁴⁵	Video-Based Anomaly Detection in Monitoring Systems Using Deep Learning <u>Diana Yermekova</u> , Botagoz Tokanova, Artem Bykov, Syrym Zhakypbekov, and Saya Sapakova	Digital Twin-Enabled Framework for Risk Assessment in Industrial IoT Network Topologies Tetiana Babenko, Kateryna Kolesnikova, Askar Askar Syssoyev, and <u>Miras Baksiyk</u>
16 ⁴⁵ –17 ⁰⁰	The Machinery Regulation (EU) 2023/1230 in the Development of Operational Safety Assurance Adam Górný and <u>Malgorzata Jasiulewicz-Kaczmarek</u>	The Use of SPC Tools in Molding Sand Control Process <u>Michał Rogalewicz</u> , Robert Sika, and Bartosz Malak

Time	Room 1	Room 2
17 ⁰⁰ –17 ¹⁵	Modernization of the Multi-operation Machine Main Drive with an Innovative V-belt Transmission Design <u>Oleg Krol</u> , Vladimir Sokolov, Oleksandr Logunov, Marin Zhilevski, and Mikho Mikhov	Integrating Artificial Intelligence into Engineering Education to Enhance Professional Foreign Language Communication <u>Aissulu Kaldarova</u> , Marco Vasquez, Anunciacion Tala Atos, Altynay Kenzhebekova, and Nurganym Amanbekova
17 ¹⁵ –17 ³⁰		Simulation Modeling of Material and Technical Resources Allocation in Agro-Processing Enterprises: Economic and Engineering Aspects Chulpan Akimbekova, <u>Olha Prokopenko</u> , Nurman Ibrishv, Altynbek Moldashev, and Vladimir Grigoruk
18 ³⁰ –21 ⁰⁰	WELCOME DINNER Venue: Restaurant “Tyubeteyka”, 32/1, Satpaev St., Almaty	

Day 3 | June 4, 2026

Time	Room 1	Room 2
8 ⁰⁰ –9 ⁰⁰	MORNING COFFEE	
9 ⁰⁰ –17 ⁰⁰	VIDEO PRESENTATIONS: https://www.youtube.com/watch?v=4nZXFmWUALE	
9 ⁰⁰ –11 ⁰⁰	KEYNOTE SESSION II Session Chair: Jan PİTEL, Technical University of Košice, Slovak Republic	
9 ⁰⁰ –9 ⁴⁰	Bridging Engineering and Medicine: Design Automation of 3D Printable Anatomical Devices for Point-of-Care Healthcare Keynote Speaker: Filip GÓRSKI, Poznan University of Technology, Poland	

Time	Room 1	Room 2
9 ⁴⁰ –10 ²⁰	From Perceiving Uncertainty to AI Reasoning: Why Decisions Still Drift? Keynote Speaker: Le LIU, Northwestern Polytechnical University, China	
10 ²⁰ –11 ⁰⁰	Navigating Augmentation and Automation in the New Industrial Revolution Keynote Speaker: Torbjørn NETLAND, ETH Zurich, Switzerland	
11 ⁰⁰ –11 ³⁰	COFFEE BREAK	
11 ³⁰ –12 ³⁰	BUSINESS TO SCIENCE Session Chair: Slawomir LUSCINSKI, Kielce University of Technology, Poland	
11 ³⁰ –12 ³⁰	Needs of Universities in Developing On-campus Training Infrastructure for University-Enterprise Integration Invited Speaker: Iurii GAIDAIENKO, Yalong Intelligent Equipment Group Co. Ltd., China	
12 ³⁰ –14 ⁰⁰	LUNCH	
14 ⁰⁰ –15 ⁴⁵	SESSION 5: ADVANCED MANUFACTURING Session Chair: Khrystyna BERLADIR, Sumy State University, Ukraine	SESSION 6: ADVANCED MANUFACTURING PROCESSES (ONLINE) Session Chair: Vera YERMAKOVA, International IT University, Kazakhstan
14 ⁰⁰ –14 ¹⁵	Towards Piezoelectric Locally Resonant Metamaterial Manufacturing with Injection Moulding David Almeida, Stefan Janssen, Raul Campilho, Francisco Silva, and Elke Deckers	Study and Analysis of Cutting Parameters in Aluminum Alloy Machining <u>Marin Zhilevski</u> , Mikho Mikhov, Krasimir Kanev, Oleg Krol, and Vladimir Sokolov
14 ¹⁵ –14 ³⁰	Improvement and Modification of the Metal Cutting Process in Tire Components <u>Raul Campilho</u> , Naiara Sebbe, Arnaldo Pinto, Marlene Brito, and Isabel Figueiredo	Statistical Analysis to Minimize Dimensional and Vibrational Perturbations in Shaft Welding <u>Edisson Calderón Freire</u> , Cristian Redroban, Fernando Pantoja, Hugo Sanchez, and Adrew Barrera

Time	Room 1	Room 2
14 ³⁰ –14 ⁴⁵	Geometric Optimization and Thickness Reduction of Metal Packaging for the Transportation of Dangerous Goods: A New Design Paradigm Naiara Sebbe, <u>Francisco Silva</u> , Marlene Brito, Ana Júlia Viamonte, Alexandra Gavina, Isabel Figueiredo, and Rita Sales-Contini	Synergistic Effects of Geometric Parameters and Material Selection on the Mechanical Performance of FDM 3D Printed Components <u>Olha Kalman</u> , Patrik Fejko, Vladimir Simkulet, Ján Pitel', and Jozef Husár
14 ⁴⁵ –15 ⁰⁰	Vibration and Wear Analysis of Milling Ti/CFRP/Ti Stacks with Primecon coating <u>Naiara Sebbe</u> , Francisco Silva, Andresa Baptista, Gustavo Pinto, and Ricardo Alexandre	Real-Time Assembly Verification in Smart Manufacturing Using a YOLOv8-Based Vision Framework <u>Myroslav Omelianenko</u> , Jozef Husár, Ján Pitel', Olha Kalman, and Justyna Trojanowska
15 ⁰⁰ –15 ¹⁵	Comparative Study of Uncoated and Primecon-Coated Carbide Tools in the Trimming of CFRP/Ti6Al4V FMLs <u>Naiara Sebbe</u> , Francisco Silva, Andresa Baptista, Gustavo Pinto, Ricardo Alexandre, Iván Iglesias and Rita Sales Contini	Decision-Oriented Evaluation of DIN 1.2738 Steel Milling Using MQL and Dry Cutting Conditions: A Multi-Criteria Approach <u>Buse Güney</u> , Alper Buğra Tarakçı, Ahmet Kayra Kayalı, Akın Gürlek, and Orhan Çakır
15 ¹⁵ –15 ³⁰	A Brief Systematic Review of Industry 4.0 Technologies and Their Applications in Advanced Manufacturing <u>Isabel Mendes Pinto</u> , Alexandra Gavina, Isabel Figueiredo, Ana Júlia Viamonte, and Arnaldo Pinto	Sustainable High-Performance Milling of Aluminum 6061-T651: A Comprehensive Comparative Analysis of Dry, MQL, and Nanographene-Reinforced N-MQL Strategies <u>Asım Goksu</u> , Ertugrul Satoglu, and Orhan Cakır

Time	Room 1	Room 2
15 ³⁰ –15 ⁴⁵	Exploring the Risks at Work: A Deep Analysis Francisco Silva, Naiara Sebbe, <u>Isabel Pinto</u> , Arnaldo Pinto, and Raul Campilho	Laser, Ultrasonic and Shot Peening Post-Processing of Additively Manufactured Inconel Alloy 718 Strengthened by HIP and Heat Treatment: Comparison of Surface Texture, Roughness and Hardness <u>Dmytro Lesyk</u> , Pradeep Menezes, Bohdan Mordyuk, Aitzol Lamikiz, and Indrajit Charit
15 ⁴⁵ –16 ¹⁵	COFFEE BREAK	
16 ¹⁵ –18 ⁰⁰	DSMIE WORKSHOP Moderator: Kateryna KOLESNIKOVA, International IT University, Kazakhstan	
16 ¹⁵ –18 ⁰⁰	AI-Augmented Agile for Industrial Management and Lean Manufacturing Invited Speaker: Bohdan HAIDABRUS, Accenture Baltics / Riga Technical University, Latvia	Smart Energy from Biomass: Thermochemical Conversion, Simulation, and Digital Tools Invited Speaker: Paweł CZYŻEWSKI, Poznan University of Technology, Poland
19 ⁰⁰ –22 ⁰⁰	AWARDS CEREMONY & GALA DINNER Venue: Restaurant “Tyubeteyka”, 32/1, Satpaev St., Almaty	

Day 4 | June 5, 2026

9 ⁴⁵ –10 ⁰⁰	BRIEF ORIENTATION
10 ⁰⁰ –18 ⁰⁰	CULTURAL PROGRAM (Outside of Almaty): RECREATION CENTER TAU-ASU
18 ⁰⁰ –20 ⁰⁰	RETURN TO ALMATY

Video Presentations

The video presentations are available online via the following YouTube stream:
<https://www.youtube.com/watch?v=4nZXFmWUALE>.

Participants may access the presentations at any time during or after the conference by opening the link in a web browser or through the YouTube application. No registration is required.

1. **Improving the Design of the Drilling Pump Piston Elements.** Vasyl Mykhailiuk, Lesya Shkitsa, and Ruslan Deineha.
2. **Modeling of Effective Operating Parameters of a Two-Piston Pulsation Homogenizer for Microemulsions.** Kyrylo Samoichuk, Volodymyr Yalpachik, Volodymyr Voitsekhivskyi, Oleksandr Napnenko, and Olha Mykhaylytska.
3. **Planning of an Experiment to Determine the Force Parameters Acting on a Carbide Cutting Tool.** Vladyslav Ruban, Vitalii Derbaba, Oleksandr Bohdanov and Olha Ostroukhova.
4. **Evaluating General-Purpose 3D Generative AI as a Preliminary Tool for Mechanical Reverse Engineering: Observations on Trellis 2 and Hunyuan3D-2.1.** Daniel Danci, Cristian Barz, and Petrica Sitar.
5. **QazNLP: A Morphology-Aware Kazakh NLP Toolkit with Structured Decoding for POS and NER.** Aigerim Aitim and Anel Auyezova.
6. **Selection of the Content for the Development of Professional Competence of Engineering Teachers.** Olena Titova, Petro Luzan, Oleksandr Koshuk, Iryna Mosia, and Vasyl Shust.
7. **Justification of the Influence of Transmission Type on Changes in Traction and Energy Performance of Tractors.** Anatolii Panchenko, Angela Voloshina, Igor Panchenko, Olena Titova, and Maksym Svyrenko.
8. **Structural and Kinematic Synthesis of a Differential Mechanism for Actuating an Inertial-Type Vibration Exciter.** Vitaliy Korendiy, Pavlo Krot, Oleh Parashchyn, Rostyslav Stotsko, and Olena Lanets.
9. **Model of Formation of Managerial Competence and Leadership Skills of Heads of Engineering Departments in the Process of Digitalization of Professional**

- Training.** Viktor Nagayev, Yuliia Sahachko, Oleksandr Hridin, Galyna Nagayeva, and Sofiia Minenko.
10. **Structural-Modal Analysis of Spherical/Cylindrical Shells Based on the Finite Element Method and Artificial Intelligence.** Viktoriya Pasternak, Maria Khomyak, Nazar Struk, Nazar Filiuk, and Oleg Zabolotnyi.
 11. **Investigation of Cutting Cross-Sectional Area Using Special Form Milling Cutters Based on a Developed Virtual Instrument in the LabVIEW Environment.** Serhii Patsera, Vitalii Derbaba, Vladyslav Ruban, Oleksandr Bohdanov, and Roman Nosov.
 12. **Formation of Secondary Structures in Surface Gradient Layers of Metallic Materials, Taking into Account Surface Defects, Impurity Diffusion, and Oxidation Reactions.** Mykola Denisenko, Olena Deviatko, Nataliia Kanivets, Mykhailo Tuziuk, and Leonid Lisovskyi.
 13. **Increasing Labor Productivity in an Industrial Enterprise in the Conditions of Single-Piece Production Using Modern Technologies and Artificial Intelligence.** Fedir Novikov, Dmytro Novikov, Valeriy Zhovtobryukh, Oleksii Yermolenko, and Yanina Chupryna.
 14. **FEA System for Lead Screws with Multi-Agent Design Optimization Capability.** Volodymyr Kopei, Oleh Onysko, Yurii Bukatka, Anatolii Verkalets, and Taras Holyk.
 15. **Modelling the Operation of an Adaptive Control System for a Brick Blank Forming Machine.** Viacheslav Perepelytsia, Oleksandr Petrov, Sergiy Petrishyn, and Serhii Melnyk.
 16. **Influence of Laser Processing Parameters on Porosity and Mechanical Properties in ABD-900AM During Laser Powder Bed Fusion.** Valentyn Riaboshtan, Serhii Badyuk, Rostyslav Podolskyi, Nataliia Pinchuk, and Mykola Tkachuk.
 17. **Energy Optimization of Batch Vibroextraction from Plant Raw Materials: Methodology and Experimental Verification.** Volodymyr Zavalov, Taras Mysiura, Nataliia Popova, Yuliia Zaporozhets, and Viktoriia Sapiga.
 18. **Using Order Statistics for Estimating Lower and Upper Size Bounds.** Nataliia Lamnauer and Oleksandr Kupriyanov.

19. **Factors Influencing the Effectiveness of Training Agricultural Engineers in the Process of Forming Their Professional Competence.** Olexandr Domin, Ganna Romanova, Leonid Romanov, Svitlana Zagurska, and Natalia Kulalaieva.
20. **Design and Validation of a Smart LMS Component Based on Microsoft Teams for Engineering Education within a Digital Smart EcoSystem.** Oleksandr Humennyi, Serhii Kubitskyi, Yuriy Kubitsky, Dmytro Humennyi, and Sviatoslav Kyiashko.
21. **Peculiarities of Magnetoresistive Properties of Medium-Entropy Alloy (FeNiCo)_xCu_{1-x}.** Oleksandr Pylypenko, Iurii Volk, Oksana Haponova, Iryna Pazukha, and Yurii Shkurdoda.
22. **Plasma-Driven Synthesis of Cu–Ni Oxide Nanoarchitectures on Copper Surfaces.** Andrii Breus, Yelyzaveta Baranova, Yelyzaveta Zasad, Semen Holovko, and Vitalii Hurianov.
23. **Improving Energy Efficiency of a Milk Jet Homogenizer.** Alexandr Kovalyov, Oleksandr Lomeiko, Vitalii Koshulko, Oleh Tertyshnyi, and Natalya Slyvka.
24. **Justification of Nozzle Parameters of a Counter-Jet Homogenizer for Emulsion Preparation.** Nadiia Palianychka, Oleksandr Lomeiko, Mykola Andrushchenko, Mykola Voinalovych, and Yurii Zhuk.
25. **Technological Assurance of High-Performance and High-Precision Machining of Materials on Finishing Operations.** Valeriy Zhovtobryukh, Fedir Novikov, Dmytro Novikov, Oleksii Yermolenko, and Andrii Ivashura.
26. **Synthesis of Rope Pulley for Elevator Winch of Parallel Structure.** Andrii Boiko, Olena Naidenko, Oleksandr Besarab, Vladislav Suvorov, and Yaroslav Sokolov.
27. **Substantiation of the Parameters of Trunnion Distributors of Orbital Hydraulic Motors in Order to Improve Their Output Characteristics.** Angela Voloshina, Anatolii Panchenko, Sergey Kiurchev, Irina Tynyanova, and Andrii Zasiadko.
28. **3D Printed Metal Parts Quality Assessment by X-Ray Diffraction.** Natalia Lishchenko, Michael Morris, Vasily Larshin, and Igor Dudarev.
29. **Investigation of The Effect of Hip Prosthesis Proximal Geometric Design on Bone Fixation and Stress Distribution Using The Finite Element Method.** Fevzi Bedir, Deyan Koçak, and Luqman Alatrushi.

30. **AI-Enhanced Mobile and Collaborative Robotic Systems for Continuous Monitoring and Automated Maintenance of Production Processes in Real Time.** Isak Karabegovic, Ermin Husak, Predrag Dasic, Volodymyr Tonkonogyi, and Liubov Bovnegra.
31. **Optimization of Combined Thin-Layer Silane Coating.** Volodymyr Komakha, Nina Merezsko, Olga Komakha, and Viktor Osyka.
32. **CleverAI Logos: Intelligent Decision Support Ecosystem in Healthcare using Machine Learning Models.** Olena Hrybiuk and Volodymyr Maksymenko.
33. **Artificial Intelligence for the Design and Optimization of the Technological Equipment: A Review.** Vitalii Kolos, Anna Burduk, Jozef Zajac, and Vitalii Ivanov.
34. **Influence of Working Fluid Temperature and Delivery Coefficient of Regulated Axial Piston Pump on Power Supplied to Hydraulic Motor.** Oleh Bereziuk, Oleksandr Petrov, Inna Vishtak, and Viktor Kolomiiets.
35. **Evaluation of Physical and Mechanical Parameters of Minced Sausage Mass With the Introduction of Transglutaminase Enzymes.** Igor Palamarchuk, Olexandr Domin, Volodymyr Voitsekhivskyi, Oleksandra Chervotkina, and Olena Prokopenko.
36. **The State of Formation of Energy Efficiency Competence in Professional Colleges of the Construction Sector of Ukraine.** Andrii Kalenskyi, Tetiana Gerliand, Nataliia Vanina, and Inna Hrytsenok.
37. **Analytical and Numerical Evaluation of Multifunctional Spatial Trusses on Certified Truck Frames.** Kostyantyn Holenko, Oleksandr Dykha, Maksym Dykha, Volodymyr Dytyniuk, and Mykhailo Burian.
38. **Improvement of the Combined Tuber Crops Cleaning Process and Development of Equipment for its Implementation.** Dmytro Dmytrevskyi, Gregoriy Deynichenko, Miushfik Bakirov, Dmytro Horielkov, and Vitalii Chervonyi.
39. **Thin Absorbent Paper from Cotton–Eucalyptus Cellulose.** Liudmyla Andriievska, Nina Merezsko, Nataliia Marchuk, Tetiana Hlushkova, and Volodymyr Komakha.

40. **Theoretical Study of the Influence of the Thermal Conductivity Coefficient of Tool Material During the Cutting of Bone Substances.** Andrii Slabkyi, Oleksandr Povstyanoy, Leonid Koval, Nataliia Imbirovych, and Bohdan Smetjukh.
41. **Performance Evaluation of a Roller Press For Dewatering Lime-Softening Sludge.** Anton Vasylenko, Maksym Lukianenko, Viktor Moiseev, Eugenia Manoilo, and Yurii Manoilo.

Keynote Speakers

Keynote Speaker



Raul Duarte Salgueiral Gomes CAMPILHO,

Ph.D. in Mechanical Engineering

Polytechnic of Porto, Portugal

He is an associate professor with habilitation of the Mechanical Engineering department of ISEP, Polytechnic University of Porto (Portugal), where he teaches several courses of the Bachelor's and Master's degrees in Mechanical Engineering. Since 2023, he has been the vice-director of CIDEM – Center for Research & Development in Mechanical Engineering, and since 2026, the vice-director of the Ph.D. in Mechanical Engineering and Sustainable Manufacturing. His research mainly focuses on adhesive joints; structural adhesives; design of bonded joints; experimental testing; Finite Element Method; Extended Finite Element Method; Meshless Methods; Cohesive Zone Models; composite materials; numerical modeling of composite materials; micromechanics; macromechanics; design of mechanical structures; flexible production; automation; robotics; and actuator systems. He was the principal investigator of 1 national R&D project and participated as a researcher in 8 national R&D projects as a researcher, all funded through competitive calls. He also participated in 1 R&D project in collaboration with the Portuguese industry (funded through competitive calls). He is/was the (co)supervisor of 10 PhD (2 concluded and 8 ongoing) and 298 MSc. theses. His publication record contains 444 articles & reviews published in Web of Science-indexed journals and 198 book chapters. He also coedited 3 books and was the (co)author of over 800 communications presented at international conferences and 18 at national conferences.



<https://www.researchgate.net/profile/Raul-Campilho>



<https://www.linkedin.com/in/raul-campilho-4a397b290>

Keynote Speech Topic

Next-Generation Product Development: Integrating Human-Centered Design, Circularity, and Simulation for Sustainable Manufacturing

The increasing pressure on industry to achieve sustainability, efficiency, and user acceptance is reshaping traditional product development paradigms. This keynote talk presents an integrated framework for next-generation product design and manufacturing that combines Human-Centered Design, Circular Economy principles, and simulation-driven optimization. By considering different case studies, the talk demonstrates how engineering design can evolve from a function-driven process into a multidisciplinary and lifecycle-oriented methodology. First, the integration of Human-Centered Design within Industry 5.0 is explored by emphasizing the role of user interaction, ergonomics, and data-driven feedback in improving product acceptance and operational safety. Next, Product Life Cycle Extension strategies are introduced as a core mechanism to enhance industrial sustainability and resilience, supported by maturity models and decision-making tools. Based on these principles, a simulation-based optimization framework is presented, which enables the redesign of industrial equipment by balancing structural performance, cost, and environmental impact through digital engineering tools. Finally, these concepts are consolidated through the development of an eco-designed collaborative robotic gripper, illustrating the practical integration of sustainability, safety, and manufacturability within a structured product development methodology. The convergence of these approaches enables the process improvement toward more sustainable, competitive, and human-centric industrial systems.

Keynote Speaker



Khamis ESSA,

BSc, MSc, PhD, CEng, FHEA, FIMechE, Director of the Advanced Manufacturing Research Group, School of Engineering,
University of Birmingham, United Kingdom

Khamis Essa is Professor of Digital Manufacturing and the Director of the Advanced Manufacturing Group at the School of Engineering, the University of Birmingham. His research is focused on advanced manufacturing technologies, for instance, metal and ceramic Additive Manufacturing (AM), Hot Isostatic Pressing (HIPing), and Incremental Sheet Forming (ISF). He has been developing novel manufacturing routes and applications for aerospace, defense, automotive, and biomedical industries. The scientific emphasis of his research is on material-process interaction. He has also been developing holistic modeling for a comprehensive understanding of process and material behaviors. His research portfolio has a research income in excess of £14M. His research has supported by major companies worldwide, including Rolls-Royce, BAE Systems, Safran, Ford, the European Space Agency, and Caterpillar. He has also been supporting small and medium-sized enterprises through major ERDF funding programs such as CASIM2 and SmartFub. Prof Essa has over 170 journal and conference publications in addition to three patents and two book chapters. Prof Essa holds two Fellowships: one from the Higher Education Academy and the other from the Institute of Mechanical Engineers. He also sets on the editorial board of three international journals.



<https://www.researchgate.net/profile/Khamis-Essa>



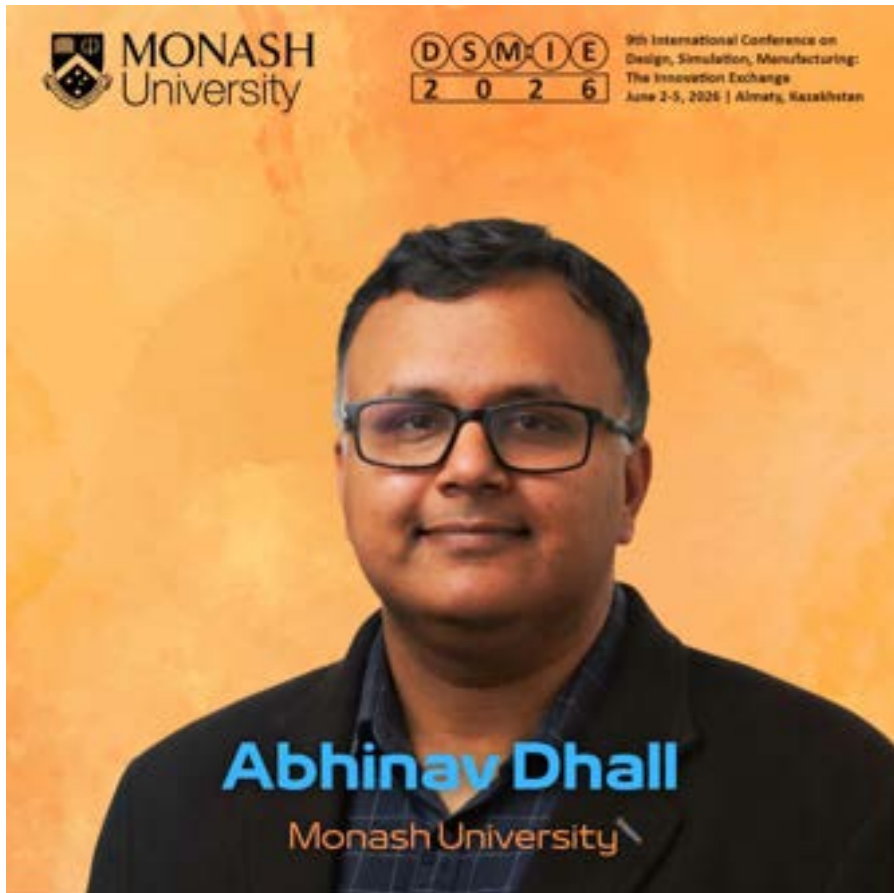
<https://uk.linkedin.com/in/khamis-essa-64a16746>

Keynote speech topic

Digital Twin as a Tool to Support the Digital and Sustainable Transformation of SMEs

The economic performance of companies in developed countries is currently being tested by the global economic situation, driven by the Covid pandemic and the tense geopolitical context. Industry 4.0 concepts clearly contribute to increasing this overall performance for large companies. But their exploitation in the context of small businesses still remains confidential for SMEs. This presentation concerns the use of sustainability combined with new technologies such as advanced robotics, artificial intelligence, and digital twins, designated by the European Commission as Industry 5.0, to solve this problem and encourage SMEs to use these concepts to improve their overall performance. The company is considered as system, and for improving its performance, the digital twin is used as an intelligent real-time decision support tool. The architecture and the development of a standardized digital twin is presented, and use cases of sustainable digital transformations of companies are exposed.

Keynote Speaker



Abhinav DHALL,

Ph.D. in Computer Science,
Monash University, Australia

Associate Professor Abhinav Dhall is the Deputy Associate Dean (International) in the Faculty of IT at Monash University. In this role, he supports the Associate Dean (International) in driving the Faculty's global engagement strategy through international research and education initiatives. Based in the Department of Data Science and Artificial Intelligence, Associate Professor Dhall's research focuses on automatic understanding of human behavior. Before joining Monash, he led the Learning Affect & Semantic Image Analysis research group at the Indian Institute of Technology Ropar. He has also held research roles at the University of Waterloo and the University of Canberra, and has been a visiting scholar at both the University of California, San Diego, and Imperial College London. He completed his PhD in Computer Science at the Australian National University in 2014 and holds a Bachelor of Technology (Distinction) from the DAV Institute of Engineering and Technology.



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Keynote speech topic

Deepfakes and Misinformation in the Age of Industry 4.0

As Industry systems become increasingly digitized and interconnected, the integrity of visual and identity-based information has become critical to maintaining digital trust. Yet the growing accessibility of generative AI-based image and video manipulation tools has made the creation of deepfakes easier than ever. This poses significant challenges for content and identity verification and can spread misinformation. In this talk, we explore multimodal approaches that are inspired from user behavior for detecting and localizing manipulations in time. A key focus of our work is on multilingual and multicultural aspects of deepfake detection. Our research draws on user studies, including those focusing on multicultural deepfakes, which provide insights into how different audiences perceive and interact with manipulated media.

Keynote Speaker



Filip GÓRSKI,

Ph.D. Eng. Professor (Assistant)

Poznań University of Technology, Poland

Filip Górski, DSc., PhD., BEng., serves as an Associate Professor at the Faculty of Mechanical Engineering at Poznań University of Technology, Poland. He leads the Division of Additive Manufacturing and Virtual Reality and acts as Vice-Dean for Science within the Faculty. His expertise lies in 3D scanning and printing technologies for medical and biomedical engineering, as well as virtual and augmented reality for both industrial and medical applications. He is also proficient in automating design processes in CAD systems and employing reverse engineering techniques. In 2013, he defended his doctoral thesis in the field of numerical simulation and experimental research of products manufactured additively on 3D printers, in 2020 he obtained his habilitation degree in the use of virtual reality techniques in mechanical engineering. Dr. Górski has been involved in numerous research projects, often in a managerial role, collaborating with industrial companies and national research centers. His international collaborations include scientists from countries such as Great Britain, Mexico, Ukraine, Romania, Slovakia, India, and Spain. He has received multiple prestigious grants and awards, notably for developing the AutoMedPrint technology, which automates the design of affordable 3D-printed prosthetics and orthotics for patients of all ages. He has authored or co-authored nearly 165 publications, including five books, and has developed several software solutions that are now implemented in industries, hospitals, and patient care facilities. He promoted over 80 engineering and master's theses related to the subject of 3D printing, virtual reality and 3D scanning, several of which have been awarded in national and local competitions.



<https://www.researchgate.net/profile/Filip-Gorski/3>



<https://pl.linkedin.com/in/filip-gorski-put>

Keynote speech topic

Bridging Engineering and Medicine: Design Automation of 3D Printable Anatomical Devices for Point-of-Care Healthcare

Personalized healthcare increasingly relies on the ability to design and manufacture medical devices tailored to individual patient anatomies. Advances in computer-aided design (CAD), reverse engineering, and additive manufacturing have created new opportunities for producing customized orthopedic devices, prosthetics, and surgical aids directly from anatomical data. However, widespread adoption of these technologies is often limited due to various factors, including the need for advanced engineering expertise. This keynote presents methodologies for design of 3D printable anatomically shaped medical devices, enabling the development of patient-specific solutions, as well as basic frameworks for design automation, helping to reduce extensive CAD skills from the involved personnel. The presented approach integrates medical imaging and 3D scanning, intelligent CAD models, and additive manufacturing workflows to create automated or semi-automated design pipelines. Such systems support the emerging Point-of-Care Manufacturing paradigm, where customized devices can be designed and produced locally in hospitals or medical centers with short response times. The lecture will discuss methodological foundations of anatomical design and design automation and illustrate them through practical case studies, including orthopedic devices, prosthetic components, and surgical planning tools. Examples from real medical cases demonstrate how automation can reduce design time, improve anatomical fit, and make personalized devices more accessible. By bridging engineering design methodologies with clinical needs, the presented approach has the potential to significantly accelerate the adoption of personalized medical technologies and bring advanced manufacturing capabilities closer to the point of care.

Keynote Speaker



Le LIU,

Ph.D., Full Professor in Design and Methods of Industrial Engineering,
Northwestern Polytechnical University, China

Dr. Le Liu earned his M.S. and Ph.D. degrees from Clemson University in the USA. He is currently an Associate Professor in the School of Computer Science at Northwestern Polytechnical University (NWPUP). In addition, he leads the Division of Visualization and Human-Computer Interaction at the Air-Space-Ground-Ocean (ASGO) National Engineering Laboratory. Before his academic career, Dr. Liu co-founded high-tech companies in Silicon Valley, Toronto, and Shanghai, where he developed cutting-edge digital human solutions powered by artificial intelligence. His research focuses on the intersection of artificial intelligence, human-AI collaboration, data visualization, and computer graphics. He has published over 30 papers in top-tier venues, including TVCG, IEEE VIS, AAAI, IJCV, and IMWUT. Dr. Liu has secured funding as the Principal Investigator from the National Natural Science Foundation of China and various leading industry partners. He was recognized as the Young Science and Technology New Star of Shaanxi Province and has received more than 10 innovation awards. He actively contributes to the research community as a chair and organization committee member of ChinaVis, the China-Japan Joint Visualization Workshop, and the CSIG-VIS International Lecture Series.



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Keynote speech topic

From Perceiving Uncertainty to AI Reasoning: Why Decisions Still Drift?

Making sound decisions under uncertainty remains a fundamental challenge across domains. The difficulty often lies not in missing information, but in the gap between how uncertainty is presented and how it is actually perceived and used in reasoning. Our work begins with empirical evidence that people systematically misinterpret uncertainty in real-world scenarios. This led us to explore a range of approaches to improve its expression, including implicit representations, redesigned visual encodings, and immersive techniques. While these methods improve understanding, they also reveal a key limitation: clearer representations do not necessarily lead to better decisions. This raises a deeper question. If perception improves, why do decisions still drift? The emergence of large language models offers a new way to examine this issue. By comparing LLMs with hundreds of human participants under the same high-uncertainty conditions, we observe a striking pattern. LLMs interpret uncertainty more accurately but still exhibit decision biases like those of humans. These findings suggest a shift in perspective, from improving representation to understanding reasoning. They point to the possibility that such biases are not merely errors, but reflect deeper mechanisms shared by both humans and AI.

Keynote Speaker



Torbjorn NETLAND,

Ph.D., Full Professor and Head of the Chair of Production and Operations Management,

ETH Zurich, Switzerland

He is the President of EurOMA, a Member of the World Economic Forum's Global Future Council on Advanced Manufacturing and Value Chains, and a Fellow of the European Academy for Industrial Management. His research on achieving and sustaining manufacturing excellence appears in leading scientific journals such as Management Science, Journal of Operations Management, Production and Operations Management, and MIT Sloan Management Review. Torbjørn is the recipient of the 2023 Jack Meredith Best Paper Award of the Journal of Operations Management, two Shingo Research Awards, and numerous teaching awards. He is co-author of the 800-page textbook "Introduction to Manufacturing: An Industrial Engineering and Management Perspective" (Routledge, 2022). Torbjørn is also co-founder of EthonAI (www.ethon.ai) – a startup offering state-of-the-art AI solutions for analytics in manufacturing. Currently, he serves as the Deputy Head of Department at D-MTEC, ETH Zurich.



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Keynote speech topic

Navigating augmentation and automation in the new industrial revolution

When it comes to Industry 4.0, after all is said and done, a lot more has been said... than done. Manufacturers are often lost in digital transformation – overwhelmed by technological opportunities in search of problems to solve. But there's a better way. In this keynote, Prof. Netland presents his group's research on how innovative technologies drive transformative improvements in leading manufacturing firms. Placed into an overarching framework, Prof. Netland explores real-world examples and insights into how new technologies, from drones to AI, are reshaping the future of manufacturing excellence. This keynote discusses the advancements toward fully automated, lights-out factories and introduces a straightforward matrix to demystify Industry 4.0 and its use-case-driven implementation.

Invited Lecturers

Invited Lecturer

RTU RIGA TECHNICAL UNIVERSITY

DSMIE
2026

9th International Conference on Design, Simulation, Manufacturing: The Innovation Exchange
June 2-5, 2026 | Almaty, Kazakhstan

Bohdan Haidabrus
Accenture Baltics, Riga Technical University

Bohdan HAIDABRUS

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Dr. Bohdan Haidabras is a strategic delivery leader and scientist with hands-on expertise in IT Project and Program Management, currently working as a Project & Program Manager at Accenture Baltics and serving as an Associate Professor at Riga Technical University. With over 17 years of experience in IT project management and more than 5 years in executive roles, he is a recognized leader in Agile delivery, digital transformation, and data program execution. Dr. Haidabras holds a PhD in Project and Program Management and an MBA in Strategic Management. He is certified as a Professional Project Manager (PMI PMP), SAFe® 6.0 Release Train Engineer (RTE), Professional Scrum Master (PSM I), and Associate CMMI. He has extensive experience in implementing leading methodologies, including PMI PfMP, IPMA, P2M, Prince2, SAFe, Scrum, and Kanban. His industry focus as a Release Train Engineer at Aztec Group includes managing data platform project portfolios and driving enterprise-wide digital and Agile transformations. In academia, Dr. Haidt has made notable scientific contributions to Agile delivery, data program management, and scaled governance models. He is the author of more than 27 peer-reviewed journal articles and conference papers on topics such as the implementation of the Scaled Agile Framework in data-driven environments, challenges of incremental execution in Agile data programs, and the role of Agentic AI in project and delivery management. His work bridges theory and practice, providing practical frameworks and empirical insights for managing complexity in modern IT organizations. He is a regular speaker and contributor to plenary sessions at international conferences, including those focused on systems engineering, digital innovation, and Agile transformation. In addition to his research, Dr. Haidabras actively mentors students, facilitates Agile workshops, and contributes to curriculum development in project management and information systems.



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Invited lecture's topic

AI-Augmented Agile for Industrial Management and Lean Manufacturing

The growing adoption of artificial intelligence (AI) in industrial environments creates new opportunities to enhance Agile practices beyond traditional implementation models. This workshop introduces the concept of AI-augmented Agile, where AI technologies support and enhance decision-making across the delivery lifecycle. AI augmentation is applied to key activities such as: flow improvement, value delivery, dependency management, data validation, and performance monitoring, enabling more scalable incremental releases.

The approach is particularly relevant for industrial management, where complex, data-intensive systems require continuous optimization. AI-augmented Agile can support productivity improvement through predictive planning, enhance quality management via automated validation and anomaly detection, and enable incremental, data-driven improvement of operational processes and case-based decision-making.

Furthermore, the workshop explores how AI-augmented practices align with Lean Manufacturing principles. AI can enhance value stream mapping and development by identifying bottlenecks and inefficiencies, support continuous improvement through real-time feedback loops, and improve flow measurement in Kanban systems using advanced analytics.

Participants will gain a structured perspective on integrating AI into Agile and Lean environments, along with practical insights into applying AI-augmented approaches into delivering high-quality incremental releases to industrial systems and manufacturing processes.

Invited Lecturer



Paweł CZYŻEWSKI

Ph.D., Associate Professor

Poznań University of Technology, Poland

Dr Eng. Paweł Czyżewski is an energy engineer and researcher specializing in advanced combustion systems, low-emission technologies, and alternative fuels, with a strong focus on ammonia- and syngas-based energy systems. He is affiliated with Poznań University of Technology, where he is an assistant professor at the Institute of Thermal Energy. He obtained his PhD in 2022 for his research on the high-temperature combustion of syngas under preheated air conditions. He also holds degree in Mechanical Engineering. He has gained international experience through research internships at the TU Wien branch (Bioenergy 2020+) in Austria and the KTH Royal Institute of Technology in Sweden, where he conducted experimental studies on biomass gasification, syngas cleaning and fuel synthesis. Dr Czyżewski's research focuses on flameless combustion, gas turbine systems, and the utilization of low-carbon fuels such as ammonia, hydrogen, and CO₂-diluted methane. He has contributed to numerous experimental and CFD-based studies on CH₄/NH₃ combustion, syngas reburning, and industrial-scale combustion systems. His publications appear in leading journals such as Energy, Fuel, and Energies, and include widely cited works on ammonia co-combustion and emissions, co-authored with researchers from Poland and abroad. He has participated in and led multiple research projects in collaboration with industry, focusing on emission reduction, combustion optimization, and renewable energy technologies, contributing to the development of carbon-neutral energy systems.



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Invited lecture's topic

Smart Energy from Biomass: Thermochemical Conversion, Simulation, and Digital Tools

This workshop provides a comprehensive introduction to the production of electricity, heat, and energy carriers through thermochemical conversion processes of biomass and solid fuels, with a focus on combustion, gasification, pyrolysis and bioogas production. Participants will explore the role of biomass and solid waste utilization in the transition toward low-emission energy systems, including their potential to reduce dependence on imported fossil fuels, improve air quality, and contribute to carbon-neutral energy generation. Special attention is given to the selection of appropriate technologies based on feedstock availability and infrastructure constraints. The workshop integrates an IT-focused component, introducing digital tools for process simulation, data analysis, and system optimization. Participants will gain insight into the use of computational modelling, real-time monitoring systems, and elements of machine learning to support efficient operation of modern energy conversion systems e.g. biogas plants.

The workshop follows an interactive, theory-to-practice methodology, combining lectures with discussion. Participants will develop skills in system analysis and design, supported by computational tools, preparing them for real-world applications in sustainable energy engineering.

Invited Speaker

Invited Speaker



Iurii GAIDAIENKO

Ph.D., Senior Research Scientist

Yalong Intelligent Equipment Group Co. Ltd

He received his education at the National Technical University of Ukraine “Igor Sikorsky Kyiv Polytechnic Institute” and defended Ph.D. thesis at the Institute of Electrodynamics of the National Academy of Sciences of Ukraine, specializing in electromechanics. Since 2017, he has been working as a Senior Research Scientist at Yalong Group. He has also served as a Visiting Professor at Zhejiang Institute of Industry and Trade (2020–2023) and Ningxia Vocational University (since 2024). Previously, he held academic positions at the National Technical University of Ukraine “Igor Sikorsky Kyiv Polytechnic Institute” as a Senior Lecturer (2017) and Assistant Lecturer (2012–2017). His professional experience also includes work as a Technical Expert at the Resource Efficiency and Clean Production Center of Ukraine (2012–2017, part-time) and ABB Ltd. Ukraine (2010–2011).



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Invited speaker's topic

Needs of Universities in Developing On-campus Training Infrastructure for University-Enterprise Integration

According to McKinsey Investigation Report, economic transformation and industrial upgrading require a sufficient number of engineers: by 2030 China's intelligent manufacturing talent gap will reach 2.4 million people, and digital twin engineers, industrial vision technicians are more than 60% of it. In the field of Digital transformation about 13 million new jobs will be created in the next five years, including data analysis, AI algorithms, and Industrial Internet of Things (IIoT) skills, the entire chain of R&D, production, and supply chain. Unlike the training bases led by traditional universities, the "Top level training base" from the perspective of the enterprise has realized the value upgrade from a "skills training site" to a "talent ecological hub". It should be: 1. "precision matcher" for talent training; 2. "innovation testing ground" for technological transformation; 3. "ecological connector" of industry and education collaboration. Using AI technology it is possible to crack the bottleneck of "high cost-risk, and low efficiency" of traditional training, and build a three-dimensional integrated intelligent system of "technical architecture + resource integration + training model". The training base should have "one platform and four centers": 1. Engineering Practice Innovation Center; 2. Technology Innovation R&D Center; 3. Digital Technology Application Center; 4. Social Training Service Center. High-level and high-quality professional development must adhere to the "four investigations and adjustments": Investigate industrial development trends and adjust professional construction; Investigate the job needs of enterprises and adjust the curriculum; Investigate and develop technology to dynamically adjust course content; Investigate the work scene of the enterprise and adjust the teaching methods.

Abstracts
Chapter I
Manufacturing Processes and Intelligent
Production Systems
Part 1
Advanced Machining and Cutting
Processes

Towards Piezoelectric Locally Resonant Metamaterial Manufacturing with Injection Moulding

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Humans are prone to developing health problems when often subjected to excessive noise, which includes, e.g., depression and sleep deprivation. Traditional noise and vibration control solutions involve, e.g., the use of heavy tuned mass dampers or the addition of large-volume absorptive material, which conflict with the trend of efficient use of material and space. Therefore, to address these problems, locally resonant metamaterials were developed, which are characterized by being lightweight and capable of creating a stop band effect, i.e., a frequency band that impedes vibration propagation. To move towards robust mass-manufacturing, in this study, the production of locally resonant metamaterials using injection moulding (IM) was researched. In which piezoelectric elements (PE) were selected to integrate the IM process, to allow the resulting IM part to possess vibration reduction properties. An existing injection moulded modular panel piece is considered as host structure in this work, where through finite-element analysis the optimal dimensions of a PE were selected to enhance vibration attenuation. Next, an insert to a modular mould was designed to over mould and integrate the PE in the host structure, integrating a connector and the required cabling. Finally, IM tests confirmed that excessive shrinkage and warpage were responsible for the fracture of the PE, and that adjusting the injection parameters to reduce shrinkage did not prevent the PE fracture. To resolve this issue, a two-shot IM process was mimicked, where the PE is fully embedded inside the host structure, resulting in successful over-molded panel pieces.

Keywords: Piezoelectric Locally Resonant Metamaterials, Injection Moulding, Piezoelectric Elements, Finite Element Analysis

Improvement and Modification of the Metal Cutting Process in Tire Components

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The ongoing imperative to optimize productivity within the automotive supply chain propels the enhancement and modernization of equipment and processes, with the objective of elevating the quality of the final product and enhancing overall manufacturing efficiency. This project, developed into a tire manufacturing company, tackles this idea by suggesting enhancements to a vital metal reinforcement cutting device. At the outset, considerable inefficiencies were identified in the process and product. The metal reinforcement exhibited persistent imperfections, including dimensional variations and excessive stress, resulting in a product that was failing to conform to the required specifications. The cutting system was modernized by introducing new mechanical elements. The main objective was to produce high-quality metal reinforcement quickly and continuously. For this purpose, the study focused on suggesting a novel cutting process that would culminate in a substantial reduction in setup times and scrap generation, enhancing efficiency. The scheme resulted in converting an obsolete process into dependable machinery with greater manufacturing capacity, providing superior product quality and reduced waste.

Keywords: Automotive Industry, Continuous Improvement, Scrap, Lean, 5S

Sustainable High-Performance Milling of Aluminum 6061-T651: A Comparative Analysis of Dry, MQL, and Nanographene-Reinforced N-MQL Strategies

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The machining of aluminum alloys, particularly the 6000 series, constitutes a critical node in the manufacturing supply chain for the aerospace and automotive fields. While Aluminum 6061-T651 offers an exceptional strength-to-weight ratio and enhanced dimensional stability, its high ductility and chemical affinity for cutting tool materials present significant machinability challenges, mostly notably the formation of Built-Up Edge (BUE) due to excessive heat generation at the tool-chip interface. This study presents an exhaustive experimental investigation and theoretical analysis of Nanographene-reinforced Nanofluid Minimum Quantity Lubrication (N-MQL) as a sustainable alternative to traditional dry machining and conventional MQL strategies, utilizing a rigorous experimental setup involving a solid carbide end mill. The impact of cutting environments on surface roughness, cutting temperature, and cutting forces across a matrix of cutting speeds (80–120 m/min) and feed rates (0.05–0.15 mm/rev) was analyzed. The experimental results indicate a transformative improvement with the inclusion of 1% wt. nanographene, yielding the lowest surface roughness (0.221 μm), cutting temperatures (267 °C), and cutting forces (49.61 N). N-MQL provides a statistically significant reduction in forces (13.5%) compared to dry cutting, reducing spindle load and energy consumption.

Keywords: Sustainable Machining, Aluminum 6061-T651, High Performance Milling.

Decision-Oriented Evaluation of DIN 1.2738 Steel Milling Using MQL and Dry Cutting Conditions: A Multi-Criteria Approach

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This study experimentally investigated the effects of machining parameters on surface roughness and energy consumption during the milling of DIN 1.2738 plastic mold steel; a material widely used in the mold industry. Experiments were conducted under both dry and Minimum Quantity Lubrication (MQL) conditions, utilizing varying feed rates, and cutting speeds. For each experiment, performance criteria were evaluated by measuring surface roughness, cutting force, and cutting temperature. The collected data revealed that MQL application significantly improved surface quality, reduced cutting temperature and force, and provided a more stable machining environment compared to dry cutting. Based on the experimental results, three sets of parameters were determined according to different production priorities: the optimum parameters that prioritize surface quality, target fast chip removal, and balance both criteria together with energy efficiency. VIKOR (ViseKriterijumska Optimizacija I Resenje) analysis, a multi-criteria decision-making method, was used to objectively select the optimal machining parameters for each condition. The obtained findings provide important contributions in terms of high productivity, cost efficiency, and sustainability in the processing of plastic mold steels.

Keywords: Surface Integrity, Sustainable Manufacturing, Plastic Mold Steels.

Investigation of Cutting Cross-Sectional Area Using Special Form Milling Cutters Based on a Developed Virtual Instrument in the LabVIEW Environment

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The technological process of restoring the contact surfaces of wheelsets is characterized by complex geometric profiles of the workpieces, wide ranges of machining allowances, and high material hardness. The magnitude of the cutting cross-sectional area is a critical factor that can lead to forces acting on the milling cutter, causing premature tool failure due to chipping or fracture. This study aims to analyze how the cutting cross-sectional area depends on the combined geometric parameters of the special milling cutter and machining conditions. The adequacy criterion of the computer simulation results with respect to real machining conditions was defined as the value of the cutting cross-sectional area of the workpiece material within the range of 0.7 to 1.0 mm². In addition, the research methodology includes the application of a graphical-analytical method, the classification-based separation of machining process factors into independent variables and parameters, and the development of an original digitized algorithmic model of a virtual instrument implemented in the LabVIEW environment. The outcomes of the study comprise an original digitized virtual instrument, through which a diagram was generated illustrating the relationship between the cutting cross-sectional area of the workpiece material and different combinations of simulated process factors, constituting the scientific novelty of this research. The practical value of the study consists in the proposed classification of machining process factors and the positive implementation of the developed virtual instrument in education.

Keywords: Computational Scheme, Graphical-Analytical Method, Cutting Element, Digit-ized Algorithmic Model.

Planning of an Experiment to Determine the Force Parameters Acting on a Carbide Cutting Tool

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The article presents the search for an optimal solution to obtain regression models - dependencies of the tangential component of the cutting force P_z on the elements of cutting parameters. The research methodology is based on the theory of experimental design, cutting theory and cutting tool. It has been established that by applying the method of planning a factorial experiment, a regression equation was obtained that makes it possible to adequately describe the dependence of such an objective function as the cutting force P_z on the main influence parameters – cutting depth t , feed rate S , and spindle rotational speed n . It is shown that the cutting force P_z is most influenced by the cutting depth t , and the lowest by the feed rate S . For example, when the cutting depth is changed in the range $t = 0.1 \div 0.7$ mm, the main component of the cutting force P_z increases by 7.59 times at feed rate $S = 1.1$ mm/rev and spindle rotational speed $n = 94$ rpm. The dependencies of planning a factorial experiment in the study of the technological process of restorative repair of the working surface of wheelsets on KZh 20 machines have been established. Using the method of planning a factorial experiment, a regression equation was obtained that allows for describing the dependence of the cutting force P_z on the cutting parameters adequately. The results of the work can be used to increase efficiency in restoring the shaped profile of the working surface of the bandages and wheelsets of rail transport vehicles without rolling out from under the locomotive.

Keywords: Wheelset, Cutting Parameters, One-Factor Studies, Cutting Force, Cutting Depth, Feed Rate, Speed.

Vibration and Wear Analysis of Milling Ti/CFRP/Ti Stacks with Primecon Coating

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The demand for new materials for use in aeronautical structures has been clear in recent years. The desire for lightweight, high-performance materials has led to the development of multi-materials, such as fibre-metal laminate composites, which are used in a variety of applications. These materials boast high mechanical strength, impact resistance, and stiffness. The machining process is significantly impacted by these characteristics due to the resultant heat generation, which leads to high rates of cutting tool wear and, in the case of fibre-metal laminate composites, layer delamination. For the machining process to be efficient, it is necessary to understand the effects of machining parameters, cutting tools, and their surface treatment. This study aims to evaluate the machinability of Ti/CFRP/Ti, considering the following parameters: cutting speed; feed; cutting length; tool wear; and vibration analysis. The study also seeks to establish relationships between these parameters. Parametric analysis identified feed per tooth (fz) as the most critical factor for tool degradation. In contrast, the investigation revealed a nonlinear effect of cutting speed (Vc): higher Vc levels led to simultaneous reductions in tool wear and vibration amplitude. Overall, Primecon-coated tools demonstrated significantly superior performance.

Keywords: Fibre Metal Laminates, Milling, Wear Behaviour, Machinability, Vibrational Analysis.

Theoretical Study of the Influence of the Thermal Conductivity Coefficient of Tool Material During the Cutting of Bone Substances

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The work conducted studies of the influence of the thermal conductivity coefficient of the tool material during the cutting of bone substances, with the possible development of a tool design with certain optimal cutting modes, which prevents the occurrence of osteonecrosis due to overheating of the bone substance. The main technical solutions for creating a tool with increased heat dissipation are adjusting the size of the contact areas between the cutting edges and bone substance, using tool materials with increased thermal conductivity coefficients, optimizing the geometric parameters of the cutting edges, using additional heat-dissipating edges and chamfers, and optimizing cutting modes to regulate the heat dissipation power. The authors also substantiated that one of the requirements for a cutting system used for medical purposes is to ensure the stability of parameters, thereby achieving a high level of ergonomics. A fundamentally new approach is shown and solve the problems of osteonecrosis during the cutting of bone tissues. To study the dependence of the thermal conductivity coefficient of the material on the heating of the cutting blade, the authors conducted modeling in SolidWorks Simulation for materials with different thermal conductivity coefficients. The results of the simulation study showed that the use of diamond as a cutting-edge material provided the greatest heat dissipation, namely 9.54 times less than the maximum blade temperature. The most promising direction for improving the process of cutting bone masses is the development of tools with a linear speed of the cutting edge exceeding 2.5 m/s and a thermal conductivity coefficient of the cutting material of at least 1000.

Keywords: Thermal Conductivity, Bone Substance, Cutting, Shavings, Heat.

Study and Analysis of Cutting Parameters in Aluminum Alloy Machining

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The selection and optimization of cutting parameters are crucial for achieving efficient, stable, and energy-efficient machining of aluminum alloys on CNC machining centers. This paper presents a combined analytical and experimental study of drilling and milling processes performed on an aluminum alloy workpiece, with a particular focus on the relationship between calculated cutting parameters and the electrical response of the machine tool drives. Analytical models are used to estimate cutting forces, torque, and power for stepped-drilling and contour-milling operations. Tool geometry, cutting conditions, and material properties are explicitly considered. The experimental studies were conducted on a CNC machining center equipped with instrumented spindle and feed drive systems, allowing the acquisition of electrical current signals during real machining conditions. The results reveal a strong correlation between analytically predicted mechanical loads and the measured drive current responses. Variations in tool diameter, chip thickness, and cutter engagement are clearly reflected in the amplitude and periodicity of the spindle and feed drive current signals. The application of peck drilling and multipass milling strategies improves process stability and reduces dynamic loading. The findings demonstrate that drive current signals can be effectively used as sensorless indicators for monitoring cutting load, process stability, and energy consumption. The proposed approach provides a practical basis for intelligent diagnostics and optimization of aluminum alloy machining processes in modern machining centers.

Keywords: Machining Center, Drive System, Machine Operations, Processing Modes.

Life Cycle Assessment Approach to Individualized 3D-Printed Orthopaedic and Prosthetic Devices

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The main objective of this paper is to conceptualize and clearly define the lifecycle of individualized medical products, based on experience gained during the development of the AutoMedPrint system and its progression to Technology Readiness Level 9 (TRL9). For this purpose, the article analyses the stipulations outlined in legislative acts and normative documents. These investigations reveal that the fundamental stage in this pursuit involves defining and analyzing the lifecycle of the medical device. Within this article, the concept of the "lifecycle of medical devices" is elucidated and clarified, on the basis of experimental studies realized during creation of the whole system. Each lifecycle stage is described in detail, forming the basis for developing, implementing, and maintaining a quality management system within AutoMedPrint to support TRL9 achievement. This lifecycle has already been successfully applied in the production of orthopedic and prosthetic devices, enabling less experienced engineers to participate and improving communication with patients. However, certain weaknesses have been identified. A novel lifecycle model is introduced, allowing for process assessment and understanding of their interrelationships. This model has been validated in studies with AutoMedPrint system patients, detailed in previous and ongoing case studies. The proposed lifecycle model is universal and can be used by laboratories and researchers to establish lifecycles for other medical devices, such as implants or orthotic devices. It provides a foundation for constructing new systems tailored to specific product needs, patient groups, or regional requirements.

Keywords: Lifecycle, Medical Device, Orthosis, Prosthesis, Regulation, Process.

Comparative Study of Uncoated and Primecon-Coated Carbide Tools in the Trimming of CFRP/Ti6Al4V FMLs

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Machining hybrid CFRP/Ti6Al4V/CFRP stacks poses significant challenges due to the disparate properties of the constituent materials, which often result in severe tool wear and vibrational instability. The present study examines the capabilities of Primecon coating on dynamic stability and wear behaviour during the trimming of hybrid stacks. Trimming tests were conducted at cutting speeds of 60 m/min, 70 m/min and 80 m/min, with feed rates of 0.03 mm/tooth and 0.10 mm/tooth. The research concentrated on the relationship between cutting forces, vibration velocity (RMS), and resulted wear. The results suggest that the Primecon coating plays a crucial role in providing tools thermal and dynamic protection. At the highest productivity regime, uncoated tools exhibited chaotic vibrational behavior. There were also high radial force fluctuations. These were due to severe titanium adhesion. Conversely, the coated tools exhibited sustained signal stability and diminished vibration amplitudes by up to 25%. A SWOT analysis showed that thermal insulation and dynamic damping are worth the cost of applying the coating for aerospace work. This work establishes a basis for enhancing the performance of stack machining, showing that nanostructured coatings are vital for preventing damage to machine components caused by vibrations during high-speed trimming processes.

Keywords: CFRP/Ti stacks; Vibration analysis; Primecon coating; Trimming; Tool wear; Cutting forces.

Technological Assurance of High-Performance and High-Precision Machining of Materials on Finishing Operations

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The purpose of the work is a theoretical analysis of the conditions for increasing the productivity and accuracy of material processing by cutting and the development of effective technological treatment processes on their basis with blade cutting tools and grinding for use in finishing operations. The possibility of a significant increase in output and machining accuracy by using high-speed cutting has been theoretically proven. We have ascertained that the greatest processing effect is achieved when the cutting coefficient is increased due to the reduction in the friction coefficient in the cutting zone with increasing cutting speed. The theoretical solutions found correspond to the experimental data obtained during high-speed milling with a carbide cutter of blind holes with a diameter of 10 mm, made in a cylinder block of GCr15 steel. Provided that the milling cutter rotation frequency >4000 rpm is reached, the deviation from the cylindricity of the hole and the roughness parameter of the machined surface R_a take the specified values. At the same time, processing productivity is three times higher compared to traditional finishing technology, which includes low-productivity operations of drilling, boring and honing the hole. This allows you to combine the previous and final operations into one and reduce machine time and costs for their execution.

Keywords: Cutting Speed, Technological Process, Grinding, Milling Cutter.

Part 2

Additive Manufacturing and Materials Processing

Synergistic Effects of Geometric Parameters and Material Selection on the Mechanical Performance of FDM 3D Printed Components

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Additive manufacturing based on fused filament fabrication (FFF), also re-ferred to as material extrusion (MEX), is widely used for producing compo-nents with complex internal geometries. This study investigates the influence of hole shape and type on the tensile strength of three-dimensional printed specimens. Four thermoplastic filaments (PLA PRO, acrylonitrile butadiene styrene (ABS), polyethylene terephthalate glycol (PET-G), and a wood-fiber composite) were evaluated using specimens with different hole configurations. Tensile testing was performed to determine maximum force, tensile strength, and elongation at break. Microscopic analysis of fracture surfaces was con-ducted to assess interlayer bonding quality and its relationship to mechanical performance. PLA PRO achieved the highest tensile strength (47.25 MPa), in-dicating suitability for applications requiring high stiffness but limited ductili-ty. PET-G exhibited lower strength (up to 28.66 MPa) with higher ductility, while ABS showed moderate strength and limited elongation. The wood-fiber composite demonstrated the weakest mechanical performance due to structural discontinuities. Rectangular and oval openings promoted more uniform stress distribution, whereas hexagonal openings reduced stiffness and increased sus-ceptibility to local failure. The results highlight the importance of combined material selection and geometric optimization to improve the mechanical relia-bility of FFF/MEX-fabricated components.

Keywords: Fused Filament Fabrication, Material Extrusion, Tensile Properties, Infill Ge-ometry, Stress Distribution, Thermoplastic Filaments, Hole Configuration, Mechanical Performance Optimization

Laser, Ultrasonic and Shot Peening Post-Processing of Additively Manufactured Inconel Alloy 718 Strengthened by HIP and Heat Treatment: Comparison of Surface Texture, Roughness and Hardness

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Post-processing of the additively manufactured aerospace alloy and composite products is an important issue today. The presence of various manufacturing defects on the surface and in the volume of additively manufactured metal structures/components leads to deterioration of surface quality and, consequently, operational properties. In this study, various advanced mechanical surface treatments were applied to enhance the surface integrity in Inconel 718 alloy manufactured by a laser powder bed fusion (LPBF) 3D technology and then post-processed by combining hot isostatic pressing (HIP) and 2-stage heat treatment. The LPBF-manufactured nickel-based alloy specimens were subjected to laser shock peening (LSP), ultrasonic impact treatment (UIT), and shot peening (SP) in order to compare the formed microreliefs on the surface, the roughness parameters, surface microhardness, and the obtained hardening depths. Inconel 718 alloy samples were laser powder bed fused (60 μm layer thickness, Renishaw RenAM 400 production system) using the optimized parameters and scanning strategy with the continuous wave laser emission mode. The results demonstrated that the type of microrelief, the amount of roughness, and the depth of hardening vary depending on the type of surface peening post-processing used.

Keywords: Additive Manufacturing, Inconel 718, Post-Processing, Hot Isostatic Press-ing, Solution Annealing, Aging, Surface Peening.

3D Printed Metal Parts Quality Assessment by X-Ray Diffraction

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3D printed metal parts are a breakthrough in the technology of their production, allowing to replace traditional multi-operation technological processes, however, on the condition of ensuring the required operational properties of these parts, which in turn depend on the internal structural-phase and physical-mechanical properties of the material. Therefore, the X-ray diffraction (XRD) method of assessing these properties considered in the paper is relevant. The possibilities of the XRD method are demonstrated using the example of studying the microstructure of Ti64 3D printed samples produced by selective laser melting (SLM) and having different initial states. An as-casting sample is also additionally studied. The phase composition, crystallite size, microstrain, and microhardness are determined for all samples. To be comprehensive, this study considers the microstructure and microhardness of the samples. The micro-structure is studied using optical microscopy and scanning electron microscope methods. It is established that successful identification of phases and quantitative determination of crystallites, the ratio of crystalline and amorphous fraction is achieved with careful sample preparation, namely, the use of grinding and polishing. The relationships between phase composition and microstructure, crystalline size and microhardness, crystallite size and microstrain are studied. XRD patterns for all samples are analyzed in terms of phase composition, peak intensity, their position, and broadening.

Keywords: X-Ray Diffraction, Selective Laser Melting, Laser Powder Bed Fusion; Ti-6Al-4V.

Influence of Laser Processing Parameters on Porosity and Mechanical Properties in ABD-900AM During Laser Powder Bed Fusion

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Maximizing the productivity of Laser Powder Bed Fusion (LPBF) for high-performance superalloys often compromises structural integrity due to processing-induced defects like porosity. This study investigated the impact of laser processing parameters on the porosity and mechanical properties of ABD-900AM, a nickel-based superalloy designed for additive manufacturing (3D printing). Using a constant laser power of 195 W, cubic samples were fabricated across three powder layer thicknesses (20, 40, and 60 μm) while varying scan speeds and hatch distances. Porosity was quantified via optical microscopy and image analysis to map the process windows for lack-of-fusion and key-hole defect regimes, which informed the selection of optimal parameters for room-temperature tensile testing in the as-built condition. The results demonstrate a wide and stable processing window for layer thicknesses of 20 and 40 μm , providing relative densities above 99.8%, while a significantly narrower window was observed for 60 μm layers. Despite the differences in porosity and productivity, all selected parameter sets yielded high room-temperature mechanical properties, with yield strength of 687–749 MPa, ultimate tensile strength of 1011–1068 MPa, and elongation up to ~50%. The results obtained confirm the excellent manufacturability of ABD-900AM by LPBF and provide practical guidelines for selecting processing parameters with an optimal balance between density, mechanical performance, and productivity.

Keywords: Laser Powder Bed Fusion, ABD-900AM, Powder, Porosity, Tensile Stress.

Part 3

Digital and AI Methods in Engineering

QazNLP: A Morphology-Aware Kazakh NLP Toolkit with Structured Decoding for POS and NER

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Kazakh is a morphologically rich agglutinative language where productive affixation increases lexical sparsity and complicates token-level sequence labeling. This paper introduces QazNLP, a modular toolkit that combines morphology-aware preprocessing with transformer-based contextual representations and structured decoding for two core tasks: part-of-speech tagging and named entity recognition. The proposed pipeline aligns word-level annotations with subword tokenization, integrates contextual encoding, and applies Conditional Random Fields to enforce globally consistent label sequences under task-specific constraints. Experimental evaluation on Kazakh NER resources and additional annotated data demonstrates improved sequence consistency compared to independent token classification, with the largest gains observed in ra-re and ambiguous forms influenced by agglutinative morphology. These findings indicate that coupling morphology-aware processing with structured decoding yields a practical, reusable baseline for Kazakh information extraction and supports reproducible experimentation for low-resource, morphologically rich languages.

Keywords: Kazakh NLP; Sequence Labeling; Conditional Random Fields; Transformer Models; POS Tagging; Named Entity Recognition; Low-resource Languages.

Evaluating General-Purpose 3D Generative AI as a Preliminary Tool for Mechanical Reverse Engineering: Observations on Trellis 2 and Hunyuan3D-2.1

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The democratization of 3D content creation through Large Reconstruction Models (LROMs) presents a novel opportunity for mechanical design workflows. This study explores the practical utility of two state-of-the-art architectures, Trellis 2 (Structured Latent Diffusion) and Hunyuan3D-2.1 (Multi-View Diffusion), for reverse engineering mechanical gears. Unlike traditional metrology studies, we focus on a qualitative assessment of the generated meshes as "starting points" for CAD modeling. A critical observation of our study is the inherent scale ambiguity of these models; because they generate geometry in a normalized unit space, they capture the relative proportions but lack absolute physical dimensions. Furthermore, while they struggle with the high-frequency sharpness required for functional involute curves, they exhibit remarkable robustness against specular reflections on shiny metal surfaces. We conclude that these AI models serve as powerful "digital sketch" tools, provided the user manually calibrates the scale in post-processing, rapidly generating reference geometry for software like Blender.

Keywords: Generative 3D Reconstruction, Mechanical Reverse Engineering, Large Re-construction Models, Geometric Metrology.

CleverAI Logos: Intelligent Decision Support Ecosystem in Healthcare using Machine Learning Models

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The CleverAI Logos system is set to transform patient screening for clinical trials by offering unparalleled efficiency, accuracy and inclusivity in clinical research. This AI-powered platform provides real-time updates on patient eligibility, reducing the time and resources needed for recruitment. Combining this with a motivated population can significantly speed up enrolment times. The possibility of creating anonymous, moderated support groups where people with similar conditions can share their experiences is also under consideration. It is crucial to identify these risks at the conceptual design stage in order to make informed architectural decisions. The application's value lies in its ability to perform AI analysis, for which users must consent to the processing of their most sensitive medical data. The system architecture focuses on securely handling medical data, ensuring scalability, and optimizing performance. The system consists of three variable modules that interact with each other. A mobile application called Logos. Care has been developed, as has an external LIS/MIS API module. This module provides an authorization API, a catalog of analyses, order creation, and result receipt. There is also an AI API module that accepts anonymized data and provides recommendations. Optimizing patient screening processes has transformed the landscape of clinical trials. This speeds up the delivery of new treatments and ensures that clinical trials are conducted with greater efficiency, accuracy, and ethical integrity. Experimental research confirms that the system's potential to improve patient care and clinical research is constantly growing, ushering in a new era of healthcare innovation.

Keywords: CleverAI Logos, Variable Models, Artificial Intelligence, Intelligent Systems, MIS, AI, Knowledge Representation Model, System Design.

Comparative Analysis of Cryptographic Algorithms for Securing IoT Systems

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This paper provides a systematic experimental comparison of three cryptographic approaches: the symmetric Advanced Encryption Standard (AES), the asymmetric El-Gamal algorithm, and a lattice-based post-quantum cryptographic scheme. The evaluation was performed on a physical ESP32 microcontroller to ensure practical relevance and reproducibility. The analysis considers key generation time, encryption and decryption latency, and overall computational overhead as primary performance metrics. The obtained results demonstrate that AES achieves the best computational efficiency, making it well-suited for resource-constrained IoT environments. The El-Gamal algorithm exhibits moderate performance, while the lattice-based post-quantum scheme requires significantly more computational resources due to the complexity of its mathematical operations. The study contributes to the practical assessment of cryptographic algorithms in embedded environments and supports informed selection of security mechanisms for ESP32-based IoT systems, while highlighting the need to further optimize post-quantum schemes for constrained hardware platforms.

Keywords: IoT, Post-Quantum Cryptography, Lattice Algorithms, AES, El-Gamal, ESP32.

Optimized FHE Library Based on Pailler's Algorithm for Secure Computing in IoT Networks

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This paper presents a lightweight fully homomorphic encryption (FHE) library based on the Paillier cryptosystem, optimized for resource-constrained Internet of Things (IoT) devices. While many existing HE frameworks focus on bit-level operations, this implementation supports direct arithmetic — including addition, subtraction, multiplication, and division — on encrypted integers. The research details a modular C++ architecture designed to minimize memory usage and energy consumption on embedded hardware. Validation was conducted using Atmel AVR microcontrollers, evaluating performance against NIST criteria such as execution velocity and power draw. Experimental results indicate that the library provides a practical balance between cryptographic security and operational efficiency, demonstrating linear complexity for addition and manageable overhead for complex integer tasks. This study addresses the gap in integer-oriented cryptographic tools, offering a deployable framework for privacy-preserving computations in distributed IoT ecosystems. The library's ability to perform division on encrypted data significantly extends the functionality of existing homomorphic schemes in real-world applications.

Keywords: Paillier Cryptosystem, Fully Homomorphic Encryption, IoT Security, Resource-Constrained Devices, Integer Arithmetic.

Part 4

Smart Manufacturing Systems and Industrial AI

Real-Time Navigation in Crowds: Using Optimized CCTV-Based Yolo for Robot Motion Planning

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The paper addresses a relevant and timely topic related to real-time navigation of autonomous robots in crowded environments using computer vision and optimized YOLO-based object detection. In this work, we introduce a novel uncertainty-aware optimization framework that extends traditional integration approaches by incorporating predictive motion modeling and cost-based trajectory optimization. The integration of visual perception, three-dimensional reconstruction, and motion planning is well structured and demonstrates practical engineering value. The proposed system is validated through quantitative experiments using benchmark datasets and real-world robotic scenarios, including evaluation of detection accuracy (mAP), navigation success rate, latency, and collision avoidance. Comparative analysis with baseline methods demonstrates measurable improvements in navigation robustness and real-time performance. The main focus is on applying computer vision and deep learning methods to generate up-to-date environmental maps, including optimizing neural network architectures for real-time navigation. An overview of existing approaches to obstacle detection, along with their advantages and limitations, is presented, and recommendations are provided for integrating these methods into intelligent mobile robot control systems. Particular attention is given to practical aspects, such as the construction of three-dimensional environmental models for navigation, dynamic obstacle mapping based on video stream analysis, and energy-efficient detection of moving objects using YOLO architectures.

Keywords: Computer Vision, Deep Learning, YOLO, 3D Reconstruction, Motion Planning, Dynamic Obstacle Mapping, Obstacle Detection, Autonomous Navigation.

AI-Enhanced Mobile and Collaborative Robotic Systems for Continuous Monitoring and Automated Maintenance of Production Processes in Real Time

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The implementation of Industry 4.0 technologies is transforming production processes towards smart manufacturing, which requires a high level of autonomy in monitoring, supervision, and maintenance of equipment in the production process in order to ensure maximum efficiency, reliability, and safety of industrial operations. A key advancement in this context is the integration of autonomous mobile robots (AMRs) and collaborative robots (cobots) enhanced by advanced artificial intelligence algorithms. This paper investigates the development and application of intelligent robotic systems that enable continuous monitoring of equipment operation in real time, with a focus on the combination of machine learning, computer vision, and predictive maintenance (PdM) models that allow timely diagnostics, fault identification, and automatic initiation of maintenance interventions. The proposed predictive maintenance (PdM) concept ensures the reduction of unplanned downtime, optimization of maintenance processes, and improvement of both the quality and efficiency of production operations. Additionally, human–robot collaboration enables more flexible maintenance tasks and an increased level of safety in industrial environments. Experimental results indicate a significant improvement in operational efficiency compared to traditional maintenance approaches. This study confirms that AI-supported autonomous mobile robots (AMRs) and collaborative robots represent a sustainable solution for future requirements of smart factories and Industry 4.0, while also opening opportunities for further progress towards Industry 5.0.

Keywords: Artificial Intelligence (AI), Mobile Robots (MR), Collaborative Robots (Cobots), Predictive Maintenance (PdM), Smart Manufacturing (SM).

Artificial Intelligence for the Design and Optimization of the Technological Equipment: A Review

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Artificial Intelligence (AI) has become a transformative force in manufacturing engineering, fundamentally changing the principles of design, optimization, and manufacturing of technological equipment. This study presents a systematic review and analysis of the latest approaches to integrating AI into manufacturing engineering, with a focus on the design and optimization of flexible fixtures and tooling systems. The study is based on a structured method that analyzes the most relevant and recent research on the application of AI in manufacturing engineering, focusing on machine learning, deep learning, optimization algorithms, reinforcement learning, and digital twins. The study has shown significant potential for AI to automate design decision-making and optimize manufacturing parameters, thereby greatly improving efficiency and quality in manufacturing engineering. The study has shown that most research on the application of AI in manufacturing engineering has reached a high level of technology readiness (TRL), ranging from 6 to 9. The study has shown significant potential to integrate AI with CAD and CAE tools to create intelligent, knowledge-based manufacturing solutions that align with Industry 4.0 concepts.

Keywords: Manufacturing Innovation, Artificial Intelligence, Flexible Fixture, Industry 4.0, Artificial Neural Network, Machine Learning, Deep Learning, Digital Twin, Intelligent Manufacturing Systems.

Real-Time Assembly Verification in Smart Manufacturing Using a YOLOv8-Based Vision Framework

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Vision-based inspection systems are increasingly applied in industrial assembly processes; however, their practical deployment is often limited by insufficient robustness, lack of temporal stability, and weak integration with higher-level control and visualization layers. This paper presents an end-to-end framework for automated assembly verification that combines deep learning-based object detection with deterministic logical evaluation and real-time visualization. The proposed solution is built around a YOLOv8 object detection model trained to recognize multiple assembly components under realistic operating conditions. To address the inherent variability of frame-based detection, a stability-oriented decision logic is introduced that aggregates detection results over time to ensure reliable component presence estimation. The validated outputs are transmitted via lightweight communication protocols and visualized using a Node-RED dashboard, providing operators with real-time feedback on assembly completeness and missing parts. Experimental evaluation demonstrates that the trained model achieves high detection accuracy across most component classes, with stable precision, recall, and mean average precision. The integration of rule-based logic significantly reduces false decisions caused by transient misdetections, particularly for visually ambiguous components. A key contribution of this work is the integration of YOLO-based object detection with a temporal stabilization mechanism and a deterministic assembly evaluation logic, enabling robust decision-making in real-time industrial monitoring scenarios. The results confirm that combining data-driven perception with deterministic evaluation enhances system reliability in practical assembly scenarios.

Keywords: Manufacturing Innovation, Artificial Intelligence, Flexible Fixture, Industry 4.0, Artificial Neural Network, Machine Learning, Deep Learning, Digital Twin, Intelligent Manufacturing Systems.

A Brief Systematic Review of Industry 4.0 Technologies and Their Applications in Advanced Manufacturing

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Industry 4.0 (I4.0) marks a transformative phase in industrial production, integrating digital, physical, and cognitive technologies to create smart, connected, and data-driven systems. This paper offers a brief review of the foundational principles and enabling technologies underpinning I4.0, including the Internet of Things (IoT), big data analytics, Artificial Intelligence (AI), machine learning (ML), robotics, and automation. Adopting a Systematic Literature Review (SLR) guided by the PRISMA framework, relevant studies were identified from the Scopus and Web of Science databases and analyzed to uncover key themes, implementation challenges, and technological interdependencies within the I4.0 ecosystem. The findings indicate that I4.0 technologies collectively enable digital transformation across manufacturing, enhancing productivity, flexibility, and sustainability through applications in smart factories, supply-chain optimization, quality control, and predictive maintenance. Nonetheless, persistent barriers such as interoperability gaps, cybersecurity threats, and workforce skill limitations impede widespread implementation. This review synthesizes existing scholarship to present an integrated understanding of how these technologies interact to drive systemic industrial innovation. The paper offers valuable insights for academics, practitioners, and policymakers seeking to leverage I4.0 to strengthen organizational resilience, competitiveness, and sustainable growth in the evolving landscape of digital manufacturing.

Keywords: Industry 4.0, Internet of Things, Big Data Analytics, Artificial Intelligence, Automation, Smart Manufacturing.

Video-Based Anomaly Detection in Monitoring Systems Using Deep Learning

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The article presents an experimental study of uncontrolled video anomaly detection based on the extended Memory Autoencoder (MemAE), evaluated on the UCSD Ped2 reference dataset. A fully reproducible pipeline was implemented, including unified data preprocessing, model training exclusively on normal frames, and evaluation of frame-level anomalies based on reconstruction error. The MemAE architecture uses an external memory module, which restricts the network to reconstruct only those patterns that were observed during training, which makes it possible to distinguish anomalies by increased reconstruction error. A comprehensive statistical analysis using span diagrams, KDE histograms, and QQ graphs revealed pronounced asymmetry and behavior with “heavy tail” in the error distribution, which justified the abandonment of classical Gaussian thresholds ($\mu \pm k\sigma$) in favor of a reliable percentile-based approach. The thresholds P_{90} , P_{95} , P_{97} and P_{99} were systematically compared using ROC-AUC, accuracy, completeness, and F1-measures. The results show a clear compromise: strict thresholds (P_{97} – P_{99}) effectively eliminated false alarms, while sensitive thresholds (P_{90} – P_{95}) provided a broader coverage of abnormal events. Overall, the data obtained confirms the practical applicability of MemAE as a flexible and customizable module for deployment in intelligent video surveillance systems for engineering purposes.

Keywords: Anomaly Detection, Video Surveillance, Autoencoder with Memory, Recon-struction Error, UCSD Ped2, Computer Vision.

The Machinery Regulation (EU) 2023/1230 in the Development of Operational Safety Assurance

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This study presents an analysis of the implementation of design guidelines set out in Machinery Regulation 2023/1230, with particular emphasis on requirements related to the safe operation of machinery and technical equipment. It was indicated that effective compliance with these requirements is achievable only through the application of adequate protective measures, properly selected in relation to the identified hazards. It was emphasized that the verification of the correctness of the implemented solutions should be conducted using a systematic risk assessment focused on eliminating hazards or minimizing their consequences. The analysis of the applicable legal regulations further demonstrated that the designed safety measures should take into account the provisions of harmonized Type B standards, which enable a presumption of conformity with essential requirements. The research findings confirmed that the proper selection and implementation of protective measures constitute a key condition for ensuring the safe use of equipment and structural compliance with current legal provisions, as well as normative requirements determining the approval of machinery for operation in diverse industrial and organizational environments.

Keywords: Operational Safety, Technical Solutions Risk, Technical Equipment, EU Machinery Safety Regulations.

Part 5

Modeling, Simulation, and Process Optimization

Fast Neuro-Symbolic Constraint Solving for Knowledge Extraction via NP-Complete Subset Sum Models

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Knowledge extraction from complicated texts typically requires reasoning under a variety of interacting constraints, offering basic hurdles for neural approaches. Many real-world extraction tasks may be easily characterized with basic NP-Complete optimization models drawn from Karp's classic set of combinatorial problems, such as Subset Sum and Set Cover. This research address-e efficient constraint-driven information extraction by proposing a fast neuro-symbolic approach that combines neural candidate scoring with symbolic con-sraint reasoning. The neural component creates and ranks candidate facts, while the symbolic layer assures consistency via NP-complete constraint formulations. To achieve scalability, the proposed approach employs approximation and pruning techniques that significantly reduce the effective search space while adhering to rigid constraints. The approach is evaluated on synthetic and real-world datasets that represent constrained extraction scenarios, and it is compared to purely neural and heuristic benchmarks. Experimental results show that the proposed neuro-symbolic framework improves constraint fulfillment and extraction consistency while using much less inference time than exact symbolic solutions. When combined with neural representations, the findings suggest that classic NP-Complete models from Karp's framework may be successfully repurposed as practical reasoning primitives in modern information extraction systems.

Keywords: Neuro-Symbolic Learning, Knowledge Extraction, Constraint-Based Reasoning, NP-Complete Optimization, Subset Sum Problem, Set Cover Problem, Decision Support Systems, Combinatorial Optimization.

Statistical Analysis to Minimize Dimensional and Vibrational Perturbations in Shaft Welding

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Transmission shafts in industrial systems operate under severe conditions of combined load and cyclic stresses, making them highly susceptible to fatigue failure. Welding repair presents a technically and economically viable alternative to replacement, although it introduces metallurgical, geometric, and dynamic challenges that can compromise component reliability. The objective of this study is to evaluate the effect of two welding repair procedures (SMAW and combined SMAW-GTAW) on the geometric and vibrational behavior of automotive alternator rotor shafts. The scientific novelty of this work lies in integrating dimensional-geometric analysis via 3D scanning with vibrational analysis as complementary tools to validate repair quality. The study adopts an experimental and comparative methodology, analyzing four shafts repaired under controlled conditions. Manual and combined welding techniques were employed, along with spark spectrometry chemical characterization, metallographic analysis, hardness measurements, and vibration monitoring before and after repair. The results show that both welding processes induce significant geometric perturbations, with the combined SMAW-GTAW process generating greater dimensional variations in shaft radius. Furthermore, vibrational analysis reveals changes in the system's dynamic behavior after repair. Collectively, the results confirm the importance of simultaneously controlling welding parameters, the resulting geometry, and the dynamic response to ensure the functionality of repaired shafts, providing useful technical criteria for industrial applications.

Keywords: Transmission Shafts, Welding Repair, Vibrational Analysis, Geometric Distortion, Mechanical Fatigue.

Using Order Statistics for Estimating Lower and Upper Size Bounds

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The paper proposes a new method for estimating the upper and lower bounds of sample size values of parts, based on determining the first and last order statistics of a bounded unimodal random variable. The method is characterized by its applicability to any unimodal distributions of sizes and does not require the use of a confidence probability in its reasoning. Distribution functions were found with the necessary condition for the existence of the mode of the last and first order statistics for any bounded random variable. Using these distributions, estimates for the upper and lower bounds of the bounded random variable were obtained. The relationship between the theoretical range and the empirical range, as determined by the sample size, allows for the required sample size to be defined based on a given accuracy. The application of the mathematical framework is illustrated with two examples of fuel equipment part sizes. The proposed method can be useful for quality control of size processing, determining the experimental strength of joints, and other applications related to processing experimental data. One of the most promising applications is the construction of control charts with regulation limits for the production process.

Keywords: Experimental Data Processing, Probability Distribution of a Random Variable, Allowable Value Limits, Sample Size.

Modelling the Operation of an Adaptive Control System for a Brick Blank Forming Machine

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The article presents the results of experimental research and mathematical modeling of synchronization methods for a hydraulic drive in an automated installation for forming brick blanks. The primary objective is to enhance the accuracy of brick formation by adapting the carriage speed to the clay block feed rate. The research methodology involves a comparative analysis of three functional control laws: linear, two-stage, and three-stage piecewise functions. It was established that linear control is insufficiently effective, resulting in dimensional errors that exceed permissible limits. A significant finding of the study is the non-linear influence of the return spring, which was experimentally identified to exert a dominant counteracting force only after a 0.02 m displacement threshold. Consequently, the optimal solution is a three-stage piecewise control law that adapts the hydraulic drive to specific phases: rapid acceleration, stable movement, and active spring force compensation. By optimizing the current amplification coefficients ($k_1 = 1.73$, $k_2 = 0$, $k_3 = 0.13$), the mean dimensional error was reduced to $0.71 \cdot 10^{-3}$ m. Experimental verification demonstrated a high correlation with simulated data. Statistical analysis confirms the system's stability and repeatability, with standard deviation values ranging from $0.040 \cdot 10^{-3}$ m to $0.106 \cdot 10^{-3}$ m across experimental sets.

Keywords: Mathematical Modeling, Adaptive Control System, Hydraulic Synchronization, Piecewise Function, Dimensional Accuracy.

Chapter II

Design, Simulation, and Energy Systems

Part 1

Mechanical Systems Design and Synthesis

Synthesis of Rope Pulley for Elevator Winch of Parallel Structure

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The applied layout and technical solutions of traditional elevator winches with worm gearboxes and two-speed asynchronous motors provide passenger comfortable movement and operation of passenger elevators even with rather limited capabilities of relay contact control devices. When implementing both methods of increasing cabin speed – by increasing engine speed or reducing the gear ratio, the developers faced a number of problems. At asynchronous motors speed, problems arose with the accuracy of the positioning of the cabin, at the same time, with a decrease in the gear ratio and an increase in the gear ratio, the worm gearbox loses its main useful quality – self-braking. When synthesizing new types of winches, the question arose about the necessary abandonment of the worm gearbox or the complete abandonment of the gearbox, as one of the ways to increase their efficiency. In this article, an innovative design has been proposed, and the parameters of a promising traction pulley for multi-motor elevator winches have been determined. The developed design of the traction pulley has a number of advantages: reduction of the speed of the drive motor can be achieved through the use of small rotating pulleys and a single-stage cylindrical gear, which will significantly increase the efficiency of the mechanical part for steady motion from 0.71 to 0.94 and at start-up from 0.45 to 0.63; the maximum tension of the rope along the arc of contact of the traction pulley is significantly reduced, providing increased durability of suspension ropes.

Keywords: Elevator Winches, Traction Pulley, Mechanical Part Efficiency, Drive Motor, Cabin Speed, Angular Velocity, Worm Wheel, Traction Pulley, Tension Force, Coverage Angle.

Structural and Kinematic Synthesis of a Differential Mechanism for Actuating an Inertial-Type Vibration Exciter

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This paper develops an analytical workflow for the structural and kinematic synthesis of a planar differential (two-DOF) internal-gear mechanism intended to actuate an inertial-type vibration exciter with controllable motion trajectories. The mechanism is first formalized as a planar kinematic chain, and its mobility is verified using the Chebyshev-Grübler-Kutzbach criterion, yielding $W = 2$ and motivating the selection of two generalized coordinates: the ring gear rotation $\alpha(t)$ and the carrier rotation $\phi(t)$. Closed-form relations are derived for the dependent unbalance orientation $\theta(t)$, and for the position, velocity, and acceleration of the unbalanced mass point E as a superposition of two circular motions. The scientific novelty lies in expressing trajectory synthesis conditions directly through phase constraints between $\theta(t)$ and $\phi(t)$, which reduce the path of E to the second-order curves (circle, ellipse, or a degenerate straight line) and provide explicit coordination laws between $\alpha(t)$ and $\phi(t)$. Numerical validation in Wolfram Mathematica software confirms the predicted regimes and illustrates practical tuning capabilities via phase parameters and the gear ratio $i = z_1/z_4$, including the advantageous case $i = 2$, enabling elliptical and rectilinear synthesis with a fixed ring gear. The results support the preliminary kinematic design of vibration exciters with trajectory and amplitude adjustability; however, direct transfer to industrial machines requires subsequent analysis of drive torque, gear loads, backlash/compliance, and prototype-based validation.

Keywords: Planetary Gear Train, Internal Gear Mesh, Mobility Assessment, Phase Coordination Law, Unbalanced Mass Trajectory, Gear-Ratio Tuning.

Modernization of the Multi-Operation Machine Main Drive with an Innovative V-belt Transmission Design

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This article considers a comprehensive study of the main drive design for a multi-operation milling machine. The structural formula and kinematic diagram for a three-stage gearbox using moving gear units are selected. A conceptual design for a drive unit layout supporting a wide range of process operations, including rough milling, is proposed. A comprehensive 3D modeling process for the drive assembly (gears, belt drive, etc.) in the integrated Creo Parametric CAD system is performed. The modelling process, based on a generative design methodology, takes into account parameters such as the range of cutting conditions and the characteristics of the workpiece. This allows for assigning multiple sets of loads, and the option to animate the deformation process is added at the output. In the Creo module, the procedure for constructing gear engagements based on a 2D conceptual layout of a basic gear pair is considered. It highlights the problem of increased loads on the gearbox shaft bearing assemblies during belt drive operation with constant belt pretension. An innovative design solution utilizing a V-belt drive by modifying the belt's cross-sectional profile, defined by circular arcs, is proposed. To substantiate this innovative solution, an analytical formula was developed to determine the key geometric parameters, including the side surface length of the modified belt profile. Based on analytical calculation have demonstrated the advantage of increasing the friction forces between the belt and pulley at the same belt pretension and, crucially, reducing the load on the drive shafts and bearings.

Keywords: Machining Center, Preliminary Design, 3D Modeling, New V-belt Solution.

Justification of the Influence of Transmission Type on Changes in Traction and Energy Performance of Tractors

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This paper substantiates the influence of tractor transmission type on changes in traction and energy performance during energy-intensive tillage operations. A simplified mechatronic model of a 100 kW wheeled tractor's hydrostatic drive and structural and functional diagrams for determining traction and energy performance characteristics for tractors with stepped (PST) and continuously variable (CVT) transmissions were developed, implemented in the Visual Simulator dynamic modeling environment. For the PST variant, the calculation was performed using four fixed gears, while for the CVT variant, operating points were selected to ensure engine operation within the zone of minimum specific fuel consumption. It has been shown that, with the same tractive effort on the hook, a tractor with a CVT develops 6–13% greater hook pull and 5–13% higher working speed compared to a tractor with a PST, while maintaining constant hourly fuel consumption and reducing specific fuel consumption. Areas of practical application for both transmission types are identified: CVT – for large farms with uneven soils and variable loads, PST – for farms with relatively stable operating modes, smaller cultivated areas, and limited service and investment opportunities.

Keywords: PST, CVT, Operational Characteristics, Structural-Functional Diagram, Trac-tion Mode, Soil.

Part 2

Computational Mechanics and Engineering Simulation

Implementation of Successive Overrelaxation Method for Solution of Elliptic Boundary-Value Problem

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This paper investigates the Successive Over-Relaxation (SOR) method as an efficient numerical approach for solving elliptic boundary value problems. The analysis begins with a brief review of Jacobi iteration theory and progresses toward the development of the SOR framework. Particular attention is devoted to matrix ordering properties and structural conditions that govern convergence behavior. Explicit criteria for selecting the optimal relaxation parameter are presented, and their influence on convergence rates is examined. The asymptotic performance of the method is analyzed in terms of computational complexity and error reduction. A numerical strategy for estimating the spectral radius of the Jacobi iteration matrix is also discussed. The theoretical results are validated through the numerical solution of the two-dimensional Laplace equation on a unit square domain under various boundary conditions. Computational experiments confirm that optimal parameter selection significantly improves convergence and reduces overall runtime. The proposed approach demonstrates that SOR remains a competitive and reliable solver for large-scale elliptic problems in scientific and engineering applications.

Keywords: SOR Method, Jacobi Iteration Fixed Point, Consistency, Similarity, Transformation, Optimal SOR Parameter, Spectral Radius, SOR Convergence Rate, Non-Dirichlet Elliptic Boundary Value Problem.

Investigation of The Effect of Hip Prosthesis Proximal Geometric Design on Bone Fixation and Stress Distribution Using the Finite Element Method

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The fixation performance of femoral components in total hip prostheses is significantly influenced by the proximal geometry, which governs stress transfer and mechanical stability at the bone-implant interface. This study investigates the biomechanical impact of various proximal geometric designs on stress distribution and fixation quality using the Finite Element Method (FEM). Three distinct analysis groups were developed based on design variables such as neck cross-section, proximal taper angle and neck-shaft angle. CAD models were generated using Siemens NX, and numerical simulations were performed in ANSYS with boundary conditions conforming to ISO 7206-4:2010 and ASTM F2996-20 standards. The results showed that designs featuring proximally tapered surfaces with angled linear textures achieved superior stress homogeneity and reduced peak stress concentrations, particularly in cortical bone regions. These findings underline the necessity of optimizing proximal surface geometry to improve implant longevity and reduce revision risk. The proposed modeling and evaluation framework offers a systematic approach for enhancing femoral stem designs in orthopedic applications.

Keywords: Hip Prosthesis, Femoral Stem, Finite Element Analysis, Implant Fixation.

Analytical and Numerical Evaluation of Multifunctional Spatial Trusses on Certified Truck Frames

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The study presents a comprehensive methodology for evaluating the strength and load distribution in a multifunctional spatial truss mounted on a certified truck frame, with emphasis on preserving the structural integrity and certification compliance of the base vehicle. A variational formulation is proposed to analytically predict the reactions in bolted joints prior to conducting resource-intensive finite element analysis (FEA). The developed mathematical model determines optimal bolt positions to minimize reactions, and its predictions for uniaxial loading scenarios demonstrate good agreement with FEA results (within 9%). A spatial frame designed for installation on the Mercedes-Benz Unimog 405 chassis was analyzed under four loading modes, including complex three-dimensional cases. The critical stress ($\sigma_{\max} = 307.93$ MPa) and deformation ($\delta_{\max} = 15.77$ mm) were observed under diagonal loading within acceptable safety margins (localized yielding length ≈ 5 mm). Notable differences in bolt reactions (up to a 5.87 ratio) emphasized the need for high-strength bolts (class ≥ 10.9) and local reinforcements. The proposed approach enables preliminary optimization of custom-mounted structures without violating regulatory certification restrictions, offering practical value for manufacturers of specialized truck-attached equipment.

Keywords: Variational Formulation, Truck-Mounted Equipment, Truss Deformation, Structural Integrity, Stress, Yield Criterion, Safety Factor, Ansys.

FEA System for Lead Screws with Multi-Agent Design Optimization Capability

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There are many studies of lead screws using FEA, but there are not many free and easy-to-use tools for effective multiparameter optimization of their design according to stress and strain criteria. Using free software, actor models and a multi-agent approach to optimization, an FEA system has been developed that is able to detect the influence of various design parameters and thread-cutting tool parameters on stresses and strains and effectively search for their optimal values. The system agents are written in Python, run in parallel using the Ray package, and employ different, complementary strategies to find the minimum, thereby increasing optimization efficiency. In particular, the system allowed to detect the dependence of the maximum equivalent stresses in the lead screw on the rake angle and the angle of inclination of the cutting edge of the thread cutter cutting this thread, and to substantiate the rational values of these parameters. Increasing the rake angle to 5° causes a more dramatic increase in these stresses compared to increasing the inclination angle to 5°. The source code of the system is published on GitHub.

Keywords: Thread Profile, Stress, Thread Cutter, Rake Angle, Actor Model.

Structural-Modal Analysis of Spherical/Cylindrical Shells Based on the Finite Element Method and Artificial Intelligence

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The article investigates the problem of computer modeling of natural modes and natural frequencies of spherical and cylindrical shells with free edges. The study considers isotropic shells with different geometric parameters (thickness, radius) and material properties (Young's modulus, Poisson's ratio, density). A comparative analysis of numerical results obtained using the Finite Element Method (FEM) and a neural operator model was performed. The results demonstrate that the Neural Operator accurately reproduces the topology of natural vibration modes and significantly reduces computation time compared with the traditional FEM approach. Numerical experiments on modeling the spectrum of natural frequencies depending on the h/R ratio and material properties made it possible to analyze the influence of geometric and physical-mechanical parameters on the dynamic behavior of shell structures. Based on these results, generalized dependencies were obtained that can be used for constructing parametric models and for rapid prediction of frequency characteristics during the design process. It was also shown that the Neural Operator ensures high accuracy in reproducing these dependencies while substantially reducing computational costs compared to classical FEM calculations. The practical significance of the research lies in the possibility of applying the proposed approach for fast prediction of dynamic characteristics of shell structures in the design of sensors, resonators, acoustic system components, and other engineering devices. The combination of FEM and Neural Operator can be considered a promising direction for the development of computer modeling methods, providing a balance between accuracy and computational efficiency.

Keywords: Prediction, Natural frequencies, Mode Shapes, Hybrid Modeling, Simulation, Intelligent Algorithms, Structural Dynamics, Young's Modulus, Poisson's Ratio.

A Cosserat Rod Approach to Static Analysis of Multi-Muscle Soft Manipulator Segments

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This study proposes a computational approach for modeling soft continuum robots driven by McKibben pneumatic muscles. The approach involves four hierarchical stages based on Cosserat rod theory and enables computations using Powell's algorithm to solve equilibrium configuration problems. This allows for the modelling of different configurations of soft elements within a segment, as well as additional external forces. A distinctive feature of this approach is its high computation speed, which is only 1.7 seconds. The paper also examines parametric modelling for varying the number of muscles (2-6) and the pressure differential (0-8 bar). Furthermore, the working volume scales sublinearly as $N^{0.65}$, indicating a decrease in efficiency with the addition of additional actuators. The pressure-displacement dependence exhibits a transition from linear ($\Delta p < 2$ bar) to nonlinear ($\Delta p > 3$ bar) behaviour, reducing actuator efficiency by approximately 25% at high pressures. These results provide actionable design guidance for pneumatic manipulator optimisation and demonstrate the suitability of this approach for rapid design space exploration prior to hardware prototyping.

Keywords: Soft Robotics, Static Equilibrium, Powell's Hybrid Algorithm, McKibben Pneumatic Muscles.

Geometric Optimization and Thickness Reduction of Metal Packaging for the Transportation of Dangerous Goods: A new design paradigm

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Packaging becomes more competitive, enabling lower sales prices. Transportation becomes easier due to reduced weight, while simultaneously mitigating environmental impact. Therefore, reducing the thickness of tinplate packaging becomes an attractive project for packaging companies. UN packaging, certified for the transport of dangerous goods, is subject to rigorous approval tests to ensure environmental and social safety during transportation. Optimizing the geometry of the packaging and its closure system is imperative. Therefore, the main goal of this work is to optimize packaging designed for the transport of dangerous goods, particularly paints, in accordance with ADR 2019, because there is a real gap in studies addressing this specific problem. Through experimental testing, the current condition of the packaging was characterized, thereby facilitating the identification of its critical failure points. The proposed improvement solutions were based on quality tools, namely the Ishikawa Diagram. Additionally, benchmarking was conducted, and to complement the quality approach, numerical analysis using ABAQUS CAE® software was employed. Validation of the study was completed with the manufacture of prototypes incorporating different solutions for the packaging's flange and top. The tests demonstrated a significant improvement in strength and ease of opening for the end customer.

Keywords: Geometric Optimiztion; Thickness Reduction; Dangerous Goods Transport; Metal Packaging; Tinplate; Environmental and economic sustainability; Natural resources saving.

Part 3

Process Engineering and Industrial Equipment

Improvement of the Combined Tuber Crops Cleaning Process and Development of Equipment for its Implementation

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The article presents an improved combined method for cleaning tuber crops, combining short-term heat treatment with excess-pressure steam followed by mechanical cleaning in a single working chamber, along with equipment for its practical implementation. The purpose of the study is to substantiate the rational parameters of the combined process of cleaning tuber crops and assess the impact of processing modes on the quality of cleaning and loss of raw materials. The proposed equipment is a horizontal rotating drum with cleaning rollers of various shapes and sizes, in which thermal, mechanical, and hydrodynamic effects on the surface of tuber crops are simultaneously implemented. Excess pressure steam is supplied directly to the working chamber, which provides intensive and controlled heating of the surface layer with minimal thermal damage to the pulp. A sharp pressure drop creates conditions for the destruction of the bond between the peel and surface tissues, after which the peel residues are removed due to friction on the rollers and washing off with water jets. The work experimentally investigated the influence of steam pressure, duration of heat treatment and mechanical cleaning on the depth of thermal exposure, peel separation force, percentage of surface cleaning and raw material loss. It was established that the optimal combination of parameters allows achieving high-quality vegetable cleaning with reduced material and energy costs. The practical value of the study lies in the possibility of using the proposed equipment at small processing plants without prior calibration of tuber crops in shape and size.

Keywords: Combined Cleaning Process, Vegetables, Steam Heat Treatment, Mechanical Post-Cleaning, Surface Layer, Raw Material Losses, Cleaning Quality, Process Parameters.

Improving Energy Efficiency of a Milk Jet Homogenizer

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Dispersing is one of the most energy-intensive operations in the overall technology of dairy product processing. The results of studies conducted by leading researchers indicate that a significant improvement in energy efficiency cannot be achieved when using the most widespread homogenizer designs. A hypothesis was proposed by a researcher, suggesting that the creation of a maximum velocity difference between phases is a key condition for obtaining a high-quality homogenized product while reducing the energy consumption of the process. This hypothesis was implemented in a jet-slot milk homogenizer. This paper focuses on improving the energy efficiency of the homogenizer through the installation of metering pumps. It was found that in the modernized design, the dependences of the drive power of the dispersion-medium and dispersed-phase feed pumps, as well as the homogenizer power, on productivity are close to linear. It was determined that, in order to ensure the required average diameter of fat globules, the rational width of the annular slot should be 0.1 mm. It was also established that, to obtain a product whose quality meets regulatory requirements, the optimal fat content of the cream should be 50%. Thus, the use of the modernized homogenizer results in approximately the same power consumption as the baseline homogenizer, while the average diameter of fat globules is reduced by 5–7%, indicating the promise of further research. The proposed homogenizer provides improved energy efficiency and meets the requirements of modern dairy processing technologies.

Keywords: Modernization, Jet Homogenizer, Cream, Power, Energy Efficiency.

Energy Optimization of Batch Vibroextraction from Plant Raw Materials: Methodology and Experimental Verification

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A comprehensive methodology for energy optimization of batch vibroextraction processes for extracting substances from plant raw materials has been developed. Based on experimental studies in the tea raw material-water system with aluminum sulfate model samples, the relationship between oscillation parameters, mass transfer intensity, and energy consumption has been established. A step-by-step engineering methodology for calculating energy costs for batch vibroextractors with flexible membranes has been developed, accounting for all components of energy costs: the movement of the working medium, the movement of the apparatus's moving elements, and overcoming hydraulic resistances. It has been experimentally demonstrated that the oscillation amplitude is the determining factor of process energy efficiency, but its effect on mass transfer has a nonlinear character with an optimal region. Critical ranges of operational parameters for batch processes have been established: frequency of 7-9 Hz at amplitudes of $15\text{-}20 \times 10^{-3}$ m provides maximum extraction rate at acceptable energy consumption levels. The developed methodology enables the design of energy-efficient extraction equipment and optimization of operating modes depending on raw material properties and extract quality requirements.

Keywords: Batch Vibroextraction, Process Intensification, Mass Transfer, Extraction Efficiency, Dissolution Coefficient, Engineering Calculation, Flexible Membranes.

Justification of Nozzle Parameters of a Counter-Jet Homogenizer for Emulsion Preparation

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The work is devoted to the issue of developing and improving the efficiency of technological equipment intended for the preparation of fine-dispersed emulsions – homogenization. At modern enterprises, the most widespread are valve-type homogenizers. However, along with the advantages of ensuring high indicators of processing quality, these devices also have a significant drawback – high energy consumption during the process. As a result of the conducted research, a promising type of homogenizer was distinguished, capable of providing a sufficiently high degree of dispersion at significantly lower energy consumption – the counter-jet homogenizer. In such units, the destruction of fat globules occurs as a result of the collision of two opposing jets. In the experimental part of the work, the obtained results showed that the highest degree of homogenization, 3.4, is achieved with coaxial arrangement of the nozzles and with a distance between the nozzles of 7–12 mm. The verification of the milk's dispersed composition, carried out through the analysis of microphotographs of the emulsion, confirmed that the average size of fat globules decreases by 19% compared with the corresponding indicator after processing in a traditional valve homogenizer. The uniformity of the emulsion's dispersed structure also increases. It has been established that the energy consumption of the homogenization process in a counter-jet homogenizer is 1.5 times lower than that of the most advanced devices. The results confirm the high potential of the counter-jet homogenization method.

Keywords: Counter-Jet Homogenizer, Nozzles, Degree of Homogenization, Dispersion.

Evaluation of Physical and Mechanical Parameters of Minced Sausage Mass with the Introduction of Transglutaminase Enzymes

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The studied process of preparing minced meat with the addition of transglutaminase enzymes allows to increase its biological and consumer value, to ensure effective balancing of the content of useful substances, which determines the relevance of the conducted research. The resulting product has a moderate cost, the modeling results allow to substantiate effective operating modes of the implementation of the studied process, determining their practical value. The article implements the application of the similarity theory and the method of "dimensional theory" to obtain patterns of changes in the main technological and physical factors in the preparation of minced meat mixture, which makes it possible to predict the influence of the studied parameters on the change in the quality of the resulting mixture. The study was conducted to determine the mathematical algorithm and its graphical interpretation, which determine the relationship, which provides dependencies between operating characteristics when forming the minced meat mixture with an enzyme admixture based on transglutaminase. As a result of these studies, it was proven that the intensity of mass exchange cyclically decreases with an increase in the content of transglutaminase in the mixture by 1%; Adding 4% of this additive increases the density of minced meat by 88%. At the same time, reducing the density of minced meat by 200 kg/m³ caused an increase in the mass transfer coefficient by 0.1 s/m.

Keywords: Enzyme Ingredients, Dimensional Theory, Similarity Numbers, Product Shear Parameters, Minced Meat Density, Mass Transfer Coefficient.

Modeling of Effective Operating Parameters of a Two-Piston Pulsation Homogenizer for Microemulsions

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The study is devoted to the analysis of technological equipment used in homogenization processes. Research findings indicate that existing equipment used for the production of finely dispersed emulsions requires further improvement aimed at lowering energy consumption during the homogenization process while simultaneously enhancing the quality of the resulting product. The conducted review demonstrated that the most promising type of homogenizer capable of achieving a high degree of emulsion dispersion at minimal energy costs is a pulsation-type double-piston homogenizer. Although the research was carried out using a natural microemulsion—cow's milk—the obtained results may also be applied to other binary dispersion systems. Emulsion breakup in pulsation homogenizers occurs due to significant pressure drops within small spatial zones generated by the pulsating motion of the piston. To determine the optimal operating parameters of the homogenizer, computer simulation was performed using the Ansys R18.2 software package. This approach made it possible to obtain hydrodynamic parameters of the emulsion flow that influence the effectiveness of dispersion. The conducted research allowed the determination of optimal operating conditions for the pulsation homogenizer, which in the future may be used to replace expensive valve homogenizers characterized by excessive energy consumption in modern industrial processes.

Keywords: Computer Simulation, Homogenization, Pulsation Homogenizer, Milk Emulsion.

Performance Evaluation of a Roller Press for Dewatering Lime-Softening Sludge

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The dewatering of sludge produced at wastewater treatment plants is a key step in the wastewater treatment process that affects environmental sustainability and economic efficiency. Lime-softening water treatment plants generate large volumes of inorganic sludge with exceptionally poor dewatering properties caused by its fine particle size, high water retention, and pronounced abrasiveness. These characteristics often lead to clogging, unstable operation, and accelerated wear of conventional dewatering equipment. This study evaluates the performance of a JD-type multi-disc roller press for continuous dewatering of lime-softening sludge under real operating conditions at a municipal water treatment facility. A two-stage chemical conditioning strategy 400 ppm metal-salt coagulant followed by 12-15 lbs/DT of high-molecular-weight cationic polymer was applied to enhance floc formation and reduce filtration resistance. The press achieved stable cake dryness with solids capture exceeding 95% and filtrate TSS values of 800-1200 mg/L. The self-cleaning intermeshing disc geometry ensured uninterrupted operation without clogging, while low rotational speeds (≤ 1.5 rpm) minimized abrasive wear and mechanical stress. Results also showed a 7-15 fold reduction in sludge volume and consistent performance across varying rheological conditions. These findings demonstrate that the multi-disc roller press provides an effective and energy-efficient solution for dewatering highly inorganic lime sludge, offering operational stability and reduced maintenance requirements. The study contributes new full-scale performance data for mineral-rich sludges and supports wider adoption of pressure-based multi-disc technologies in modern sludge management systems and environmentally friendly wastewater treatment technologies.

Keywords: Lime-Softening Sludge, Mechanical Dewatering, Multi-Disc Roller Press, Inorganic Residues, Chemical Conditioning, Coagulant-Flocculant Treatment, Sludge Volume Reduction, Solids Capture Efficiency, Water Treatment, Dewatering Technology, Sludge Processing.

Part 4

Fluid Power Systems and Energy Efficiency

Influence of Working Fluid Temperature and Delivery Coefficient of Regulated Axial Piston Pump on Power Supplied to Hydraulic Motor

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Analysis of available scientific sources revealed the absence of multifactorial regression models describing the power supplied to a hydraulic motor depending on such hydraulic drive operating parameters as working fluid temperature and delivery coefficient of a regulated axial piston pump. This determined the relevance and direction of the conducted research. The article establishes an adequate multifactorial regression model describing the dependence of power supplied to the hydraulic motor on the operating parameters of a hydraulic drive with a regulated axial piston pump. The obtained model can serve as a basis for creating a methodology for engineering calculations of hydraulic drive parameters. Analysis using Student's criterion showed that among the investigated factors, working fluid temperature has the greatest influence on the power supplied to the hydraulic motor, while the delivery coefficient of the regulated axial piston pump affects it to a lesser extent. A response surface of the objective function – power transmitted to the hydraulic motor – was constructed, and its two-dimensional cross-section in the plane of influencing factors was performed. This made it possible to visually demonstrate the nature of power dependence on individual operating parameters of the hydraulic drive with a regulated axial piston pump. An expression was obtained for determining the optimal working fluid temperature value for different delivery coefficient values of the regulated axial piston pump, characterizing it the condition at which the power supplied to the hydraulic motor will be maximum. The obtained dependencies will improve the efficiency of hydraulic drives from regulated axial piston pumps.

Keywords: Hydromotor Power, Working Fluid, Delivery Coefficient.

Improving the Design of the Drilling Pump Piston Elements

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The paper discusses methods for increasing the durability of cylinder-piston pairs in drilling pumps. The design features of serial pistons are analyzed, in particular the shape and parameters of sealing collars, the method of their fastening to the core, and the influence of ring projections on the stress state of elastic elements. The patterns of contact stress distribution in the sealing zone are established and the most loaded areas of the sleeve are determined. It has been shown that the main friction and wear occur on the supporting part of the sealing element, which justifies the need for its structural unloading. The expediency of using trapezoidal plastic rings, which ensure self-centering, compensation of gaps and reduction of contact stresses, has been substantiated. An improved collapsible piston design is proposed, enabling the reuse of metal parts and the replacement of only worn elements. Using the SolidWorks Simulation software package, simulation modeling of the stress-strain state was performed, and the optimal angle of inclination of the backing ring was determined, which ensures uniform distribution of contact pressure. The results obtained confirm the effectiveness of the proposed design and its prospects for increasing the service life of drilling pumps and reducing operating costs.

Keywords: Contact Pressure, Simulation Modeling, Stress-Strain State, Structural Optimization.

Substantiation of the Parameters of Trunnion Distributors of Orbital Hydraulic Motors in Order to Improve Their Output Characteristics

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The article considers the problem of substantiating the geometric and kinematic parameters of trunnion distributors of low-power bearingless orbital hydraulic motors used in mechatronic modules of self-propelled machinery hydraulic drives. The design features of the trunnion distribution system, which combines the functions of a hydraulic fluid distributor and a plain bearing, are analyzed. A calculation scheme and mathematical apparatus for determining the cross-sectional area of the distribution windows of the moving and fixed distributors are developed. The influence of the geometric parameters and kinematic schemes of the distribution system on the output characteristics of the hydraulic motor is studied using simulation modeling. It was found that when using a 5/4 kinematic scheme, the average cross-sectional area of the trunnion distribution system is 14 mm², which is 21% greater than when using a 5/6 kinematic scheme - 11 mm². Moreover, distribution systems with a 5/6 kinematic scheme have a flow area oscillation amplitude of 2.5 mm², while those with a 5/4 kinematic scheme have a cross-sectional area oscillation amplitude of 5 mm². The obtained results allow us to substantiate the rational parameters of trunnion distributors and can be used in the design of energy-efficient or-bital hydraulic motors for mechatronic modules of hydraulic drives for self-propelled vehicles.

Keywords: Kinematic Diagram, Flow Area, Mechatronic Module, Plain Bearing.

Chapter III
Materials Engineering, Digital
Transformation, and Human Factors in
Engineering
Part 1
Advanced Materials and Surface
Engineering

Thin Absorbent Paper from Cotton– Eucalyptus Cellulose

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This study substantiates the feasibility of using cotton cellulose as an alternative fibrous material for the production of thin absorbent paper. Cotton cellulose is characterized by high whiteness (80–90%), low lignin content, and the ability to form strong interfiber bonds, ensuring a combination of mechanical strength, softness, and high absorbency. Based on a comparative analysis of the morphological characteristics of fibers from various plant sources, the advantages of cotton fibers for sanitary and hygienic products are demonstrated. The composition of the paper pulp based on bleached cotton cellulose (36–46 °SR) and bleached sulfate eucalyptus wood pulp (17–23 °SR) was experimentally investigated, and the optimal component ratio of 42–48% and 52–58%, respectively, was established. Under these conditions, thin 32 g/m² paper was obtained, exhibiting increased bulk (4.23–4.27 cm³/g) and capillary absorption (36–41 mm in 10 min) while maintaining sufficient strength and stable whiteness (84–86%). Renewed cotton cultivation in Ukraine may support the development of a national raw-material base, reduce reliance on imports, and encourage environmentally sustainable paper production.

Keywords: Cotton Cellulose, Absorbent Paper, Eucalyptus Cellulose, Bulk, Capillary Absorption.

Plasma-Driven Synthesis of Cu–Ni Oxide Nanoarchitectures on Copper Surfaces

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Plasma-assisted synthesis offers unique opportunities for fabricating metal-oxide nanostructures under non-equilibrium conditions, yet the correlation between plasma spatial distributions and resulting oxide morphologies remains insufficiently understood. In this work, we investigate the growth of Cu–Ni oxide nanostructures on copper substrates exposed to a nickel hollow-cathode discharge in oxygen atmosphere. After plasma treatment, the nickel hollow-cathode structure was removed and the exposed copper surface was examined using scanning electron microscopy (SEM) and energy-dispersive spectroscopy (EDS). The results reveal a continuous but strongly nonuniform oxide layer whose morphology evolves radially in direct correspondence with the local plasma flux. At the periphery of the discharge footprint, the surface is decorated with vertically aligned filamentary nanostructures 8–15 μm in length, exhibiting aspect ratios up to 80. At intermediate radii, these structures become coarser, partially fused, and less anisotropic, indicating enhanced ion bombardment. Near the discharge axis, the oxide collapses into a dense, fine-grained Ni–Cu oxide film composed of 80–200 nm agglomerates, consistent with thermally driven coarsening under maximal plasma exposure. EDS analysis confirms the formation of mixed NiO–CuO compositions, demonstrating that nickel erosion from the hollow cathode participates actively in nanostructure growth. The results establish a direct link between radial plasma nonuniformity, ion-energy flux, and the dimensionality and texture of Cu–Ni oxide nanostructures, offering a pathway for plasma-engineered nanoarchitecture design in sensing, catalytic, and energy applications.

Keywords: Metal Oxide Nanostructures; Hollow-Cathode Plasma; Oxygen Discharge; Ion-Assisted Growth.

Formation of Secondary Structures in Surface Gradient Layers of Metallic Materials, Taking into Account Surface Defects, Impurity Diffusion, and Oxidation Reactions

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The reliability and durability of machinery depend on the preservation of materials' bulk and surface strength over time when subjected to loads and external environmental factors. Surface degradation processes consist of a large number of microscopic destruction events, resulting in the gradual reduction in the material's volume wear. The aim of this work is to investigate the secondary structure formation and defect detection in metallic materials under abrasive friction. Under friction conditions, secondary structures become saturated with oxygen, resulting in a complex phase composition. Friction loading induces deformation across multiple structural levels and extends to significant depths, contributing to the development of a thick secondary structure layer that has a high concentration of deformation defects. Owing to their high susceptibility to degradation, secondary structures may result in a friction surface that is only partially covered by such structures. For many types of contact interactions under external friction, the predominant processes are abrasive. The greatest intensity of friction surface damage is caused by abrasive particles with a small radius of curvature (sharp-edged particles), ranging in size from 0.2 mm to 2.0 mm. At all stages of friction loading, interactions occur between activated surface layers and the active components of the external environment (adsorption, diffusion, chemical reactions).

Keywords: Dissipative Structures, Abrasive Friction, Oxide Film.

Optimization of Combined Thin-Layer Silane Coating

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A combined thin-layer protective coating consisting of a biocidal adhesive sub-layer and a polyalkylhydride-siloxane-based protective top layer for cellulose-containing materials, such as paper packaging, was investigated. To effectively control the properties of the two-layer organosilicon coatings, a series of model formulations was developed and studied using an experimental design approach, which enabled regression-based processing of the obtained data and the creation of mathematical models relating material properties to coating composition. The model formulations, prepared with varying ratios of key components according to a central composite rotatable design, were evaluated for wetting angle, moisture absorption, tensile strength, and fungal resistance. Using response surface plots, a compromise region for the optimal component ratio in both the adhesive sublayer and the protective top layer was identified. It was determined that applying an adhesive sublayer containing 2–3 wt.% potassium methylsiliconate and 1–2 wt.% copper–zinc aminomethylsilicate complex, followed by a protective layer consisting of an aqueous dispersion of 10–15 wt.% polyalkylhydridsiloxanes, ensures achieving the targeted performance characteristics.

Keywords: Cellulose Materials Protection, Properties Modeling, Multi-Criteria Optimization.

Peculiarities of Magnetoresistive Properties of Medium-Entropy Alloy (FeNiCo)_xCu_{1-x}

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This work reports on the structural and magnetoresistive properties of medium-entropy thin-film alloy (FeNiCo)_xCu_{1-x} with a non-equiatomic composition. The thin-film alloys with total thicknesses of 30–100 nm were fabricated by thermal annealing of multilayer [Fe_{0.5}Ni_{0.5}/Co/Cu]_n structures at 700 K under ultra-high vacuum conditions. Electron diffraction and transmission electron microscopy studies revealed that all thermally stabilized films with Cu content up to 50 at.% are polycrystalline with grain sizes of 10–50 nm and exhibit a single-phase fcc solid solution structure with a lattice parameter of $a = 0.357\text{--}0.358$ nm. Anisotropic magnetoresistance was observed for all investigated samples, with magnitudes ranging from 0.1% to 1% depending on the Cu content, film thickness, and measurement temperature. The longitudinal and transverse magnetoresistance increased monotonically with film thickness, reaching maximum values of 1.0% and 1.2%, respectively, at a Cu content of 10% and a thickness of 80 nm. The magnetoresistance decreased by a factor of 2–5 as the temperature increased from 120 to 400 K, which is attributed to the reduction in saturation magnetization and magnetic anisotropy. The dependence of the anisotropic magnetoresistance on Cu content exhibited a nearly linear decrease with increasing non-magnetic component fraction.

Keywords: Thin-Film Alloy, Non-Equiatomic Alloy, Crystal Structure, Size Dependence, Annealing.

Part 2

Industrial Digitalization and Data-Driven Systems

An Integrated Framework for Digital Advertisement and Fleet Capacity Optimization in Long-Haul Logistics

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Inefficient use of truck capacity remains a persistent challenge in long-haul road freight transportation, leading to higher operating costs and missed revenue opportunities. This study proposes a novel integrated decision framework that coordinates sponsored search advertising with the capacitated vehicle routing problem to enhance truck utilization and improve both logistics and commercial performance. A multi-objective mixed-integer linear programming model is developed to jointly minimize unused vehicle capacity, optimize advertising expenditure, and stimulate demand along targeted delivery routes. The model incorporates a time-sensitive event score to identify regions with high demand potential and dynamically adjusts advertising intensity according to available transport capacity. A real industrial case from a leading Saudi Arabian air-conditioner company is used to validate the approach, supported by a real advertising campaign execution using Google Search Ads. The results indicate substantial reductions in unused truck capacity, including more than 40% improvement during low-demand periods and a reduction from 42.25% to 29.55% during high-demand periods, together with an approximately 20% increase in advertising return on investment. These outcomes confirm that aligning digital advertising decisions with transportation optimization can significantly reduce empty running, enhance fleet utilization efficiency, and strengthen business performance in competitive freight distribution environments.

Keywords: Integrated Logistics–Marketing Optimization, Sponsored Search Advertising, Demand Shaping in Freight Transport, Capacity Utilization, Mixed-Integer Linear Programming, Vehicle Routing Problem.

Simulation Modeling of Material and Technical Resources Allocation in Agro-Processing Enterprises: Economic and Engineering Aspects

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This study examines the effectiveness of hybrid simulation in optimizing material and technical resource allocation in agro-processing enterprises under unstable production conditions. The research addresses variability in raw-material supply, rising energy costs, ageing equipment, and unpredictable failures. A hybrid model integrating discrete-event simulation, system dynamics, and optimization is developed to capture both operational disruptions (queues, breakdowns, repair cycles) and long-term resource dynamics (inventory, maintenance funds, energy use). The model is applied to Kazakhstan, Estonia, Germany, and Poland for 2020–2024. The results show consistent performance improvements across all countries. Germany achieved a reliability index of 0.91 by 2024 through predictive maintenance and high levels of digitalization. Kazakhstan reduced energy consumption from 118 to 108 kWh per ton through improved maintenance planning. Estonia demonstrated steady reductions in queue time, reflecting effective digital monitoring. Poland improved maintenance cost efficiency and cycle stability through modernization and better scheduling. Cross-country comparison confirms that higher digital maturity, stable energy systems, and proactive maintenance policies accelerate efficiency gains. The findings indicate that hybrid simulation is an effective decision-support tool for enhancing reliability, reducing costs, and improving production stability. Future research should integrate climate risks, energy shocks, and machine-learning-based adaptive optimization to further improve model accuracy and strategic applicability.

Keywords: Hybrid Simulation, Discrete-Event Systems, System Dynamics, Production Scheduling.

GenAI in Research Idea Lifecycle: Conceptual Framework

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The emergence of generative artificial intelligence (GenAI) has fundamentally transformed scientific research processes. While existing frameworks focus primarily on traditional academic research stages, they overlook the broader innovation ecosystem from conception to commercialization. This study addresses the gap between current AI capabilities and their potential to enhance the entire research lifecycle, extending beyond conventional research to include grant acquisition, intellectual property protection, and technology transfer. We developed a comprehensive conceptual framework that systematically maps GenAI applications across all lifecycle stages through a six-phase methodology: task identification, GenAI applicability assessment, AIAS classification, ethical and integrity analysis, final task selection, and validation mechanisms. The framework incorporates contextual factors such as researcher experience, resource constraints, and environmental specifics while ensuring compliance with responsible AI use principles. The resulting framework demonstrates GenAI's transformative potential for enhancing traditional research activities and complex knowledge transfer and commercialization processes. Key contributions include systematic guidance for optimal GenAI integration points, standardized evaluation approaches across diverse lifecycle phases, and mechanisms for continuous improvement as GenAI capabilities evolve. This comprehensive approach bridges the gap between academic discovery and practical application, ultimately contributing to more efficient innovation eco-systems that maximize research investment returns and enhance societal impact through accelerated knowledge transfer.

Keywords: Research Idea, Lifecycle, Framework, GenAI.

Digital Twin-Enabled Framework for Risk Assessment in Industrial IoT Network Topologies

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Industrial IoT environments have grown into complex ecosystems where devices communicate across network boundaries that blur traditional separations between operational technology and enterprise systems. Conventional risk assessment treats vulnerabilities in isolation, assigning static scores regardless of where a device sits within the network topology. This approach loses information that matters. A vulnerability in a gateway bridging critical segments poses different risks than the identical vulnerability in a peripheral sensor, yet static scoring cannot distinguish between them. This paper proposes a framework for representing IIoT infrastructure as a heterogeneous attack surface graph, where devices and services form distinct node types linked by edges that encode connectivity, execution dependencies, and exposure pathways. A graph neural network learns to propagate risk information via message passing, producing node-level compromise probabilities that reflect both local vulnerabilities and topological context. An illustrative scenario demonstrates that topology-aware inference yields different risk rankings than CVSS alone, with network segmentation reducing compromise probability by up to 38% for topology-dependent devices.

Keywords: Cyber-Physical Systems, Vulnerability, Propagation, Graph Neural Networks, Attack, Graph Modeling, Industrial IoT, Security Threat Propagation Analysis.

The Use of SPC Tools in Molding Sand Control Process

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Driven by global competition, companies must improve process quality to better meet customer expectations. Statistical process control (SPC) is a quality management method that monitors process stability and capability through control charts and capability indices. While widely used in industrial practice, it is less commonly used for complex and highly variable processes, such as foundry processes. This article describes the successful implementation of SPC tools for molding sand quality control in an iron foundry. Previously, this control was conducted in a simplified manner using a check sheet, and it was a significant challenge for process operators to recognize when the process was no longer compliant and was affected by disturbances resulting from special causes. According to the presented methodology, $\bar{x}$ -R control charts were developed and the capability indices C_p and C_{pk} were calculated for five basic molding sand properties: moisture, compressive strength, tensile strength, permeability, and compactibility. It was presented that the speed of detecting disturbances affecting the process increased after the changes were implemented.

Keywords: Capability Indices, $\bar{x}$ -R Control Chart, Foundry Processes, Moisture.

A Cognitive Digital Triplet Architecture for Anomaly Diagnosis and Operator-in-the-Loop Decision Support in Discrete Manufacturing

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This paper proposes a laboratory-informed cognitive digital triplet architecture for discrete manufacturing systems, designed to extend conventional digital twin functionality with composite-state monitoring, causal diagnosis, and operator-in-the-loop decision support. The study addresses the need for more interpretable and adaptive digital representations of production systems that can not only mirror the physical process but also support explanation-oriented diagnosis and corrective action assessment under disturbance conditions. The proposed architecture integrates a physical reference environment based on the CP Factory modular cyber-physical production system with a FlexSim simulation model used for experimentation, calibration, and scenario-based validation. Its core analytical contribution is a composite signal layer that aggregates selected production-state variables into an interpretable synthetic indicator of system health, enabling earlier and more robust line-level anomaly detection. This layer is complemented by a causal diagnosis module that links deviations in the composite signal and local process variables with candidate root causes and possible corrective actions. An operator-in-the-loop mechanism is then used to incorporate expert judgment into the diagnostic and decision-support process. The research methodology combines laboratory data acquisition from CP Factory with simulation-based experiments in FlexSim under multiple disturbance scenarios, including processing-time variations, resource degradation, transport delays, and buffer-related disruptions. The evaluation framework considers detection quality, diagnosis usefulness, response interpretability, and selected production performance indicators such as throughput, cycle time, and system stability. The paper presents a structured cognitive digital triplet architecture, a composite-signal-based monitoring concept, and a laboratory-informed validation approach for explainable, adaptive decision support in discrete manufacturing.

Keywords: Digital Triplet, Digitalization, Industrial Growth.

Part 3

Industrial Management and Sustainable Transformation

Evaluation of Investment Attractiveness in the Context of Industry 4.0/5.0 Transformation

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The transformation of industrial systems under Industry 4.0 and the emerging Industry 5.0 paradigm has significantly changed the determinants of investment attractiveness, reducing the explanatory power of traditional financially driven evaluation frameworks. This study aims to reconceptualize investment attractiveness by integrating financial and non-financial dimensions related to digitalization, sustainability, and human-centric development. To this end, an integrated Investment Attractiveness Index for Industry 4.0/5.0 (IAI₄₋₅) is proposed, combining financial readiness, digital and artificial intelligence maturity, technological flexibility, sustainability, ESG orientation, and human-centric organizational capital. The study employs a quantitative, indicator-based methodology that includes normalization, composite index construction, correlation analysis, threshold-based necessary condition logic, and cluster analysis. The results reveal substantial heterogeneity in investment attractiveness and demonstrate that digital maturity and sustainability orientation act as structural prerequisites rather than optional enhancements. Financial readiness alone is insufficient to ensure high investment attractiveness, while human-centric organizational capabilities amplify the returns from digital and technological investments. The findings contribute to investment and industrial transformation literature and provide a practical tool for evaluating readiness for Industry 4.0/5.0 transformation.

Keywords: Composite Index, Digital Maturity, Sustainability, ESG Orientation, Human-Centricity.

Product Management Capabilities in a Data Programme Implementation

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Data Programme implementations increasingly demand a shift from project-based delivery toward product-centric value creation, particularly in environments characterized by high uncertainty, evolving business needs, and complex data architectures. This paper explores Product Management capabilities as a key enabler of effective Data Services programme implementation operating within a Lean-Agile, SAFe-based environment. The study focuses on Product Management at the Agile Release Train level, where Product Managers lead product strategy, vision, and roadmapping across one or multiple data products, such as Power BI Dashboards, API-based Data Services, Data Distribution Services, and a Data Platform itself. In complex data programmes, several Product Managers often collaborate to cover platform components, shared services, and business-facing data products. From the RTE viewpoint, Product Management plays a critical decision-support role by bridging customer needs, market dynamics, and organizational capabilities, while enabling coordinated execution across Agile teams. Based on qualitative insights from increment planning and execution in a real-world Data Services programme, the paper identifies recurring challenges, including unclear product ownership, misalignment between data platform enablement and business outcomes, and fragmented decision authority. The findings highlight Product Management capability patterns—such as outcome-oriented roadmapping, evidence-based prioritization, and continuous feedback integration from Agile teams—that improve ART-level flow, dependency management, and value realization.

Keywords: Data Based Decision Making, GenAI, Agentic AI, Agile.

Increasing Labor Productivity in an Industrial Enterprise in the Conditions of Single-Piece Production Using Modern Technologies and Artificial Intelligence

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The purpose of the work is to provide a theoretical and practical justification for the directions of increasing labor productivity and reducing the cost of manufactured products at an industrial enterprise in conditions of unit production compared to conditions of mass production. A mathematical model for determining the shortest time for manufacturing products has been developed using the example of metalworking. It has been theoretically established and experimentally confirmed that the time for technical preparation for manufacturing a product in unit production is significantly longer than the time for its manufacture. It is shown that modern CNC metal-cutting machines of the "machining center" type can be effectively used to reduce the time of technical preparation for the manufacture of a product. The possibility of significantly reducing time and costs in processing conditions with modern foreign carbide cutting tools with wear-resistant coatings has been established. Based on this, a technology for high-performance machining of complex-shaped working surfaces of 2,200 mm diameter tire mold sectors (elements) was developed on a CNC machining center. Compared to traditional machining on a milling machine, this technology allows for an 11.5-fold increase in labor productivity. Machine time was reduced from 180 hours to 15.5 hours. The time for technical preparation for product manufacturing decreased by 3.2 times and amounted to 50 hours. Practical recommendations are provided for further improving the efficiency of unit production.

Keywords: Processing Time, Product Cost, Industry 4.0, CNC Machine Tool.

Market Readiness for Data Science as a Service: An Empirical Multi-Phase Process Assessment

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The exponential increase in global data volumes compels organizations to adopt data-driven decision-making, yet many lack the necessary internal expertise. Data Science as a Service (DSaaS) offers a value to clients by allowing the externalization of specific workflow stages. This study empirically investigates the suitability of five core Data Science process phases for an outsourcing model. Based on a quantitative survey of professionals (N=83), we evaluate each phase regarding suitability, efficiency potential, integration complexity, and privacy concerns. The findings indicate significant heterogeneity across phases. The phases Model and Analyze as well as Extract Features are regarded as highly appropriate due to their strong potential for standardization. In contrast, the early phase Frame the Problem and the final phase Present Results and Deploy Code demonstrate low potential for outsourcing, mainly due to their reliance on specific contextual factors and privacy concerns. Additionally, the findings indicate that organizations handling large amounts of data tend to view DSaaS much more favorably. We conclude that DSaaS is a valuable strategic instrument for efficiency, provided it is applied selectively to specific, standardized process phases rather than as a universal solution. This paper is based on the findings of the master's thesis titled "Demand Analysis of Data Science as a Service".

Keywords: Outsourcing, Cloud Computing, Machine Learning.

A Structured Investigation of Barriers to Reverse Logistics Implementation for Post-Consumer PET Bottles

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This study investigates the barriers to implementing reverse logistics (RL) in the PET bottle industry in Sri Lanka, with a focus on identifying and analyzing the critical factors that impede its adoption. Using a combination of the Delphi method and Interpretive Structural Modeling (ISM), the research identifies eleven key barriers, which include weak government policies, inadequate infrastructure, financial constraints, and poor stakeholder collaboration. These barriers are systematically analyzed and structured to understand their interdependencies and their influence on the effectiveness of reverse logistics. The findings reveal that government policy deficiencies are the primary driving force, exacerbating other issues such as logistical costs, material contamination, and lack of public awareness. A MICMAC analysis further classifies the barriers based on their driving and dependence power. The study emphasizes the need for policy reforms, improved infrastructure, and enhanced stakeholder collaboration to address these challenges. The research provides a practical framework for policymakers, industry stakeholders, and environmental agencies to prioritize interventions that support sustainable waste management and circular economy practices. By addressing foundational barriers, the study offers valuable insights for improving reverse logistics systems and enhancing the recycling and management of PET bottles in Sri Lanka and potentially in other developing countries.

Keywords: Reverse Logistics, PET Bottles, Barriers, ISM, Sri Lanka, Circular Economy.

Navigating the Complexities of Responsible Sourcing in Global Supply Chains

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Responsible sourcing (RS) has become an important practice to incorporate environmental, social, and governance (ESG) in the global supply chain. Nevertheless, it is still very difficult to realize because of the multi-tier supplier networks, regulatory diversity, and the competing interests of the stakeholders. This paper presents a systematic literature review (SLR) of 96 peer-reviewed articles published in an original sample of 619 Scopus-indexed articles. The analysis reveals the following critical dimensions of complexity such as risk ambiguity, ethical inconsistency, stakeholder fragmentation, poor performance measurement, limited collaboration, and low adaptability. The study introduces the new framework called “RESPOND” which includes Risk Identification, Ethical Compliance, Stakeholder Alignment, Performance Metrics, Operational Transparency, Negotiation & Collaboration, and Dynamic Adaptation. This framework is used as a structured conceptual framework in dealing with complexity in responsible sourcing practices. The results have a contribution to academic literature as well as practical applications because they give a comprehensive picture of RS challenges and present a strategic way of addressing them. This study is valuable to cross-industrial insights on responsible sourcing and supports organizations to shift towards sustainable and transparent supply chain operations.

Keywords: Responsible Sourcing, Sustainable Procurement, Global Supply Chains, Sustainability, Ethical Compliance, Complexity Mitigation, Systematic Literature Review, RESPOND Framework.

Exploring the risks at work: a deep analysis

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Working is something inherent to any human being, contributing to their livelihood, or that of their family, and contributing to global development, as well. However, any activity involves risks, which can have varying degrees of severity for the worker. These risks, when translated into health problems or accidents, lead to worker absenteeism, which can have implications for the company's normal workflow. Furthermore, it always results in a drop in economic performance, which does not benefit the overall economy. While, in the past, risks to workers were poorly considered and resulted in serious problems for workers, but not always for companies (depending on the labor market supply), workers have greater access to information and preparation for work and have begun to conduct their own risk assessments and avoid jobs where the risks are higher. This results in higher costs for companies, as there are fewer potential candidates for the necessary roles. Employers, faced with this situation and aware of increased employee information, have made significant progress in reducing employee risk. This chapter aims to provide an overview of selected risks and to show how research has contributed to their identification and mitigation.

Keywords: Risk Analysis, Industrial Manufacturing, Economic Efficiency.

Part 4

Engineering Education and Human Capital Development

Factors Influencing the Effectiveness of Training Agricultural Engineers in the Process of Forming Their Professional Competence

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The article examines the main factors that affect the effectiveness of training engineers for the agricultural industry. Given their direct impact on the quality of the process of forming professional competence of agricultural engineers, the above-mentioned factors were considered, starting with the analysis of their definition and typification and ending with generalized conclusions regarding their expedient division according to classification features and the priority of the impact on the effectiveness of forming professional competence of specialists in the specialty "Agroengineering". During the study, approaches were developed to determine the priority of the studied factors and to involve a sufficient number of students of the specialty "Agroengineering" from six different educational institutions in Ukraine to assess the importance of each factor. The specified group of respondents was involved in determining the key professionally important personal qualities necessary for an agricultural engineer as a specialist. As an effective method of final processing of the study results, the Pareto diagram was used, which clearly demonstrates the priority of internal factors, and also indicates the need to increase attention to each of the specified factors while providing recommendations on the appropriateness of its development level.

Keywords: Influencing Factors, Professional Training, Higher Education, Professional Pre-Higher Education, Agricultural Engineer, Professional Competence.

Design and Validation of a Smart LMS Component Based on Microsoft Teams for Engineering Education within a Digital Smart EcoSystem

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This paper presents the design and empirical validation of a Smart Learning Management System (LMS) component implemented on Microsoft Teams and embedded within a Digital Smart EcoSystem for engineering education. The study explores how collaborative digital technologies can be systematically integrated to support the training of future engineers through structured, adaptive, and data-driven learning processes. Particular attention is given to the organization of learning content that accommodates individual learning trajectories, aligns with professional engineering requirements, and supports the development of digital engineering competence. Within this framework, the proposed Smart LMS component facilitates the development of essential engineering capabilities, including analytical reasoning, problem solving, and digital literacy, while also supporting reflective learning and professional self-regulation. By integrating analytics-driven feedback mechanisms, collaborative workspaces, and scenario-based instructional design within Microsoft Teams, the system creates a flexible learning environment aligned with contemporary engineering practice. Empirical results indicate that integrating the Smart LMS into a Digital Smart EcoSystem leads to measurable improvements in learner participation and task-oriented engagement and highlights the potential of LMS-centered, analytics-supported architectures for engineering education in the context of digital transformation.

Keywords: Smart EcoSystem, Learning Management System, Microsoft Teams, Engineering Education, Digital Learning, Smart Learning Technologies.

Integrating Artificial Intelligence into Engineering Education to Enhance Professional Foreign Language Communication

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This study examines the integration of artificial intelligence in engineering education to enhance professional foreign language communication among university students majoring in Software Engineering. Using a quantitative research design, the study involved sixty intermediate-level Software Engineering students enrolled in the Profession-Oriented Foreign Language (Intermediate) course during the first semester of the 2025–2026 academic year. The instructional intervention was implemented over a 15-week period, during which AI-supported tools were incorporated into self-directed learning tasks completed outside the classroom to support the development of foreign-language speaking skills relevant to the students' field of study. The effectiveness of the AI-based instruction was evaluated through a pre-test/post-test design. Statistical analysis of the results demonstrated a significant improvement in students' overall speaking competence following the intervention. The findings indicate that artificial intelligence can support foreign language development in engineering education by providing structured, consistent, and interactive learning opportunities. This study contributes to engineering education research by providing empirical evidence of the effectiveness of AI-supported language instruction for Software Engineering students and by highlighting the potential of AI-driven approaches to enhance professional communication skills beyond the classroom.

Keywords: ChatGPT, English Language, Speaking, AI-Assisted Learning, Technology, Skills.

The State of Formation of Energy Efficiency Competence in Professional Colleges of the Construction Sector of Ukraine

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The article presents an overview of the formation of energy-saving competence based on the analysis of the educational and professional program developed in accordance with the Standard of Professional Pre-Higher Education: Educational and Professional Degree – Professional Junior Bachelor, Specialty 192 “Construction and Civil Engineering”, as well as the educational program for the profession “Tiler”, developed on the basis of the State Educational Standard for profession 7132 “Tiler”. A questionnaire survey was used to determine the state of formation of energy efficiency competence in institutions of professional pre-higher and vocational education in Ukraine. At the ascertaining stage, the levels of formation of energy efficiency competence across all components are predominantly at a medium level. To increase the levels of formation of energy efficiency competence at the formative stage across all components, it is necessary to implement a methodology for forming this competence in future construction specialists. The methodology for forming energy efficiency competence of future construction specialists is a system of structured knowledge about principles (general and specific), content (educational components aimed at forming energy efficiency competence), methods (explanatory-illustrative, reproductive, heuristic, research-based, exercises, game-based, project-based, situational, testing, etc.), forms (problem-based and binary classes, presentation or project-based classes, simulation and game-based classes, training sessions, electives, clubs, independent work, webinars, etc.), and pedagogical technologies (project-based learning, problem-based learning technology, simulation and game-based learning, training technologies, inter-active learning, case method, contextual learning) for forming the studied quality. The proposed methodology includes the following interrelated stages: value-motivational, formative-technological, and reflective-corrective.

Keywords: Pedagogical Conditions, Environmental Competence, Sustainable Development.

Model of Formation of Managerial Competence and Leadership Skills of Heads of Engineering Departments in the Process of Digitalization of Professional Training

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The purpose of the article is to develop and theoretically substantiate a structural-functional model of digitalization of the development of leadership skills of managers, focused on the systematic improvement of their managerial competence. The research methodology is based on theoretical generalization, systemic analysis, modeling, and application of competency and system approaches. Based on an analysis of the management practices of heads of engineering departments at an agricultural enterprise, a step-by-step organizational and economic model for developing leadership competencies of engineering managers is proposed, structuring the key stages of personnel management and professional development. The developed model consists of interrelated blocks: goal-oriented, content-based, technological-process, and result-evaluation. The component structure of the formation of leadership skills of heads of engineering departments is determined, and the levels of their professional readiness are characterized (motivational-cognitive, operational-activity, and creative-developmental components). The structure of digital tools for the development of managerial competence and leadership skills in heads of engineering departments is substantiated (digital learning platforms, analytical information systems, virtual learning environments, simulation-based decision-making algorithms, and case-based communication methods). The practical value of the study lies in the possibility of implementing the proposed model into the system of engineering education and continuous professional development of heads of the engineering sector, thereby increasing the effectiveness of digital technologies and the quality of management within the framework of the enterprise's engineering services.

Keywords: Model, Leadership, Digitalization, Managerial Competence, Heads of Engineering Departments, Professional Training, Competency-based Approach, Digital Technologies.

Selection of the Content for the Development of Professional Competence of Engineering Teachers

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Continuous technological progress and rapid innovation in engineering demand systematic professional development of teachers in technical colleges. This study proposes a technique for designing the content of the development of an engineering teacher's professional competence tailored to engineering educators. The approach integrates fundamental principles, including humanistic orientation, scientific rigor, consistency, historicism, connection to practice, individualization, and accessibility. Criteria for content selection ensure practical relevance, alignment with institutional resources, consideration of prior teacher training, and incorporation of international best practices. The technique involves systematic steps which allow the process of teachers' professional development to be purposeful: analysis of teachers' needs and best practices, content selection, structuring and sequencing, and integration of technical and pedagogical knowledge. Experimental application demonstrated significant improvements in teachers' competence, encompassing teaching, methodological, digital, ecological, inclusive, and subject-specific skills. Findings suggest that a structured, principle-driven content design can effectively enhance engineering teachers' professional growth, support applied learning, and prepare them for modern challenges in engineering education. The study highlights the potential to replicate this approach across diverse institutional contexts to foster innovative, practice-oriented teacher development.

Keywords: Engineering Education, College Teacher, Teacher's Professional Development, Content Design, Competency-Based Approach.

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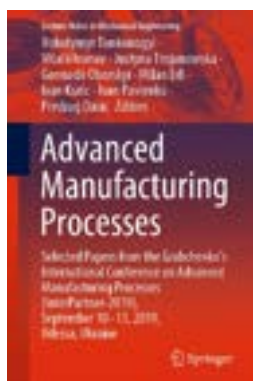


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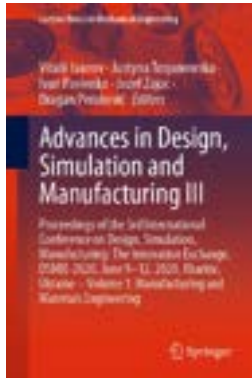


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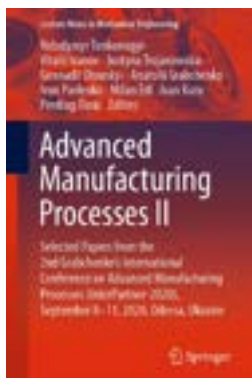


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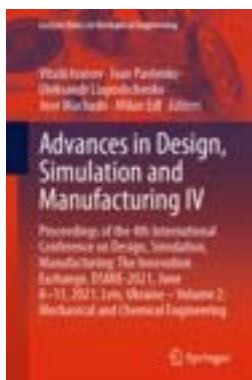


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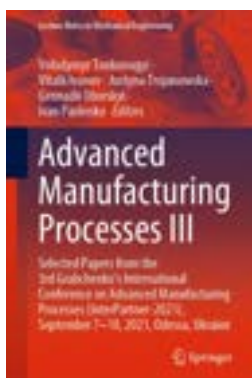


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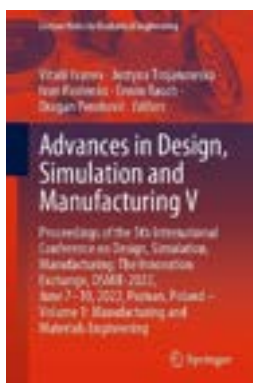


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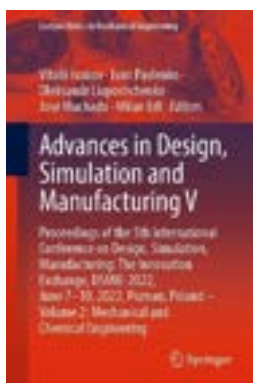


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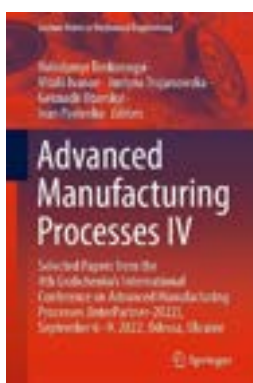


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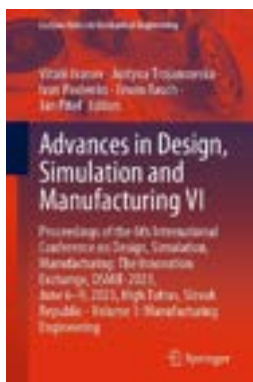


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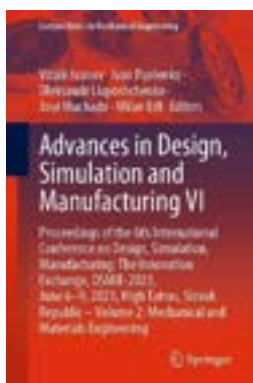


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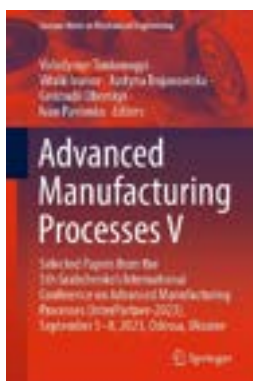


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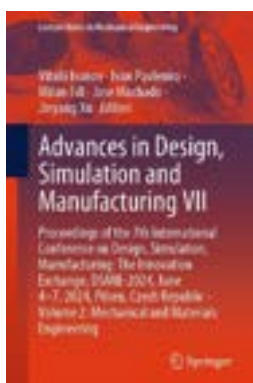


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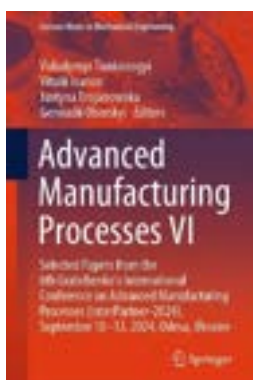


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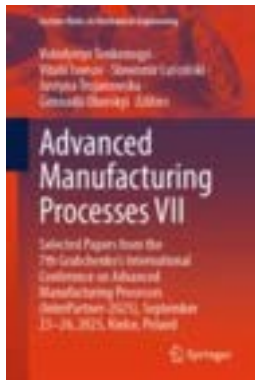


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